

In addition to the extra computing involved, the difficulty with carrying back values is that special care must be taken not to double until enough back values have been developed from the last doubling or from the initial starting.

Most programs contain provisions for both doubling and halving the interval and use two constants C_d and C_h to test

$$|p_{n+1} - c_{n+1}|$$

When halving is called for, the newly computed numbers at $n + 1$ are thrown away, the interval is halved by interpolation, and the step is tried again. If, however, halving is called for on the first predictor-corrector step, then the whole start must be redone.

Care should be taken to keep

$$C_h > 100C_d$$

Otherwise there is the risk that, because of chance fluctuations, the machine may halve and double on successive steps.

Another point should be made: If machine errors are at all likely, then a *sudden* jump in $p_n - c_n$ is probably due to machine error rather than too large a step. The above method, which throws away the step effectively, eliminates the machine error at the cost of halving and later doubling. A special test could be used to watch for sudden jumps, and when jumps are found, the step can merely be repeated.

23.12 NUMERICAL VERIFICATION

A popular approach to the problem of selecting a method for numerically integrating some differential equation is to use a computer to solve a number of special cases (often chosen because they have known solutions). While this approach is sometimes necessary, it is seldom satisfactory.

In order to check the *theory* just developed, a large number of solutions to three equations

Equation	Initial condition	Analytic solution
$y' = y$	$y(0) = 1$	$y = e^x$
$y' = -y$	$y(0) = 1$	$y = e^{-x}$
$y' = -2xy^2$	$y(0) = 1$	$y = \frac{1}{1+x^2}$

were computed for $0 \leq x \leq 10$ for various step sizes h . These equations were selected because the first two have essentially the "local linearized behavior" of the general equation and clearly reveal instability troubles. The third has a solution which is often troublesome to approximate by polynomials.

Only methods whose predictor and corrector had the same coefficients for the function values (y 's) were used, except that the classic Milne's method was also tried.

Because the evaluations of the derivatives were so easy, flat random noise was added to simulate a higher level of roundoff noise at this point *only* for half the runs.

The results, which are too voluminous to reproduce here, can be summarized as follows:

- 1 The stability conditions were verified. The use of the modifier based on $p_n - c_n$ accentuates an oscillation due to instability, but use of $p_{n-1} - c_{n-1}$ helps to prevent this tendency toward instability.
- 2 For moderately accurate solutions the use of the modifier and final-value steps helped by reducing errors by a factor of about 10 in size (provided that stability conditions were observed); that is,

$$\frac{1}{10} |p_{n+1} - c_{n+1}| \approx |m_{n+1} - y_{n+1}|$$

Whenever the step size and hence the errors were too small, then these two steps did not help, since roundoff dominated the small correction effects.

- 3 For the equation $y' = y$ the three-eighths rule, in spite of its large truncation error, did very well.
- 4 If we exclude the unstable methods, then on an average, the Adams-Bashforth, two-thirds, and one-half methods appeared not to differ significantly, with perhaps some advantage to the first.

These results should not be taken as definitive, and further study of special situations will probably reveal much more. The field of numerical integration of differential equations is a very complex one, and a complete study of it does not belong in an introductory course. Hence we shall drop the topic of numerical verification without, we hope, giving the idea that it is unimportant.

24

SPECIAL METHODS OF INTEGRATING ORDINARY DIFFERENTIAL EQUATIONS

24.1 INTRODUCTION AND OUTLINE

The predictor-corrector methods of the previous chapter require special methods until enough old values are built up to start the predictor. Probably the most widely used starting methods are those of Runge-Kutta. While these methods can also be used to compute each step, the machine efficiency is likely to be low. The typical Runge-Kutta method requires four evaluations of the derivatives for each step forward, while the predictor-corrector methods require one or two per step, and at the same time both methods have about the same truncation error. The Runge-Kutta methods often do not contain within themselves any measure of the accuracy of the solution at each step; hence it is difficult to find out when to halve or double the interval size. The idea of using two separate integrations at different step sizes has been proposed, but it is so obviously expensive of machine time that the proposal need not be taken seriously.

Whenever the system of equations of a frequently repeated problem has a special structure, it is worth examining the equations to see if one can take advantage of the special structure to reduce the amount of computation required. For example, second-order linear differential equations (with variable coefficients)

have enough special structure that special methods significantly reduce the amount of computing, when compared with the time needed to write the second-order equation in the canonical form of a pair of first-order equations and then solve them in their general form.

A great many special methods have been invented, and since this is an introductory text, we can examine only a few of them here; thus we shall merely show how a couple of special methods can be constructed. The derivations illustrate the same general ideas as were used in the previous chapters and again show the power of the general methods. The reader should, at this point, be able to construct the special cases he needs to solve his own special problems.

24.2 RUNGE-KUTTA METHODS

There are many variants of the Runge-Kutta method, but the most widely used one is the following: Given

$$y' = f(x, y)$$

$$y(x_n) = y_n$$

we compute in turn

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

This process may be described in geometric terms. At the point (x_n, y_n) we compute the slope k_1/h and using it, we go one-half step forward and examine the slope there. Using this new slope (k_2/h) , we again start at (x_n, y_n) , go one-half step forward, and again sample the slope. Using this latest slope k_3/h , we again start at x_n, y_n , but this time we go a full step forward where we examine the slope k_4/h . The four slopes are averaged, using weights $\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6}$, and using this average slope, we make the final step from x_n, y_n to x_{n+1}, y_{n+1} . If $f(x, y)$ did not depend on y , then the averaging would produce Simpson's formula. The method has an error term proportional to h^5 .

There are variants on the method as to the locations where the samples are taken and hence what weights are assigned to them in the various steps. There are also lower-order methods which use fewer samples per step, as well as higher-order methods.

It is evident that the method throws away all old information and begins each complete step anew, and hence is hardly likely to be as efficient as methods which take advantage of old information. It is also evident that there is no check on whether the step size is too small or too large, although perhaps a study of the k_i might give a clue; this is often not done. Among other proposals to monitor the accuracy are: (1) comparing k_4 of one step with k_1 of the next step, and (2) using the quantity

$$\frac{\sum_{i=1}^N (k_3 - k_2)_i^2}{\sum_{i=1}^N (k_2 - k_1)_i^2}$$

where N is the number of equations.

The method of deriving these equations does not seem to be used elsewhere in numerical analysis, and so it is not worth giving here.¹ The general spirit of the derivation is that the functions $f(x, y)$ which are on the right-hand sides are all expanded in series in powers of h , and the corresponding derivatives are equated to eliminate the lower powers of h . The result is that the method is exact for polynomials of degree 4 (or less), but is *not* equivalent to a power-series expansion about the origin, as may be seen by solving the equation (using $h = 1$)

$$\begin{aligned} y' &= \sin^2 2\pi x \\ y(0) &= 0 \end{aligned}$$

since the calculated solution will be identically zero, but the true solution is

$$y = \frac{4\pi^2}{3} x^3 + \cdots$$

24.3 SECOND-ORDER-EQUATION METHODS WHEN y' IS MISSING

Many special situations may be investigated, and in this section we shall confine ourselves to a common situation where we have second-order differential equations with no first derivative present. Frequently, if the derivative is present, a simple transformation will remove it. We shall use the same methods for investigating this problem as we used in Chap. 23.

We begin with the corrector for integrating the differential equation

$$y'' = f(x, y)$$

¹ See Kopal [29] and Ralston [49].

which we shall assume in the form

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h^2(b_{-1} y_{n+1}'' + b_0 y_n'' + b_1 y_{n-1}'' + b_2 y_{n-2}'')$$

The method of the last chapter would suggest saving two parameters, but a brief trial indicates that using a_2 as the only parameter will take us through the stability problem safely. Thus we can make the method exact for $1, x, \dots, x^5$. The results are given in Table 24.3.

The method in the last column has a zero error term; hence it has higher accuracy than the others. But the multiple characteristic root means that the propagated error ε_n will behave as

$$\varepsilon_n = C_1 + C_2 n + C_3 n^2$$

which is not pleasant to contemplate. In the other cases we have

$$\varepsilon_n = C_1 + C_2 n + C_3 (a_2)^n$$

The multiple root $\rho = 1$ is unavoidable if we make the form exact for 1 and x , and the linear growth of the error comes from an error in y' (which does not appear explicitly), propagating linearly with n .

The advantage of the two zero coefficients in the choice [39] $a_2 = 0$ makes this formula very attractive.

$$y_{n+1} = 2y_n - y_{n-1} + \frac{h^2}{12}(y_{n+1}'' + 10y_n'' + y_{n-1}'') - \frac{h^6 y^{(6)}(\theta)}{240}$$

This can also be written in the form

$$y_{n+1} = 2y_n - y_{n-1} + h^2 \left(y_n'' + \frac{\Delta^2 y_{n-1}''}{12} \right) - \frac{h^6 y^{(6)}(\theta)}{240}$$

The symmetry makes it easy to check that $G(s) \neq 0$ for $-h < s < h$.

Table 24.3

	$a_2 = -1$	$a_2 = -\frac{1}{2}$	$a_2 = 0$	$a_2 = \frac{1}{2}$	$a_2 = 1$
$a_0 = 2 + a_2$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3
$a_1 = -(1 + 2a_2)$	1	0	-1	-2	-3
$a_2 = a_2$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
$b_{-1} = \frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$
$b_0 = (10 - a_2)/12$	$\frac{11}{12}$	$\frac{13}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	$\frac{13}{12}$
$b_1 = (1 - 10a_2)/12$	$\frac{11}{12}$	$\frac{13}{12}$	$\frac{10}{12}$	$-\frac{9}{12}$	$-\frac{11}{12}$
$b_2 = -a_2/12$	$\frac{1}{12}$	$\frac{1}{24}$	0	$-\frac{1}{24}$	$-\frac{1}{12}$
$E_6 = -3 + 3a_2$	-6	$-\frac{9}{2}$	-3	$-\frac{3}{2}$	0
$\rho_1, \rho_2, \rho_3 = 1, 1, a_2$	1, 1, -1	1, 1, $-\frac{1}{2}$	1, 1, 0	1, 1, $\frac{1}{2}$	1, 1, 1

A convenient predictor to go with this corrector (we do not derive it here) is

$$y_{n+1} = 2y_{n-1} - y_{n-3} + \frac{4h^2}{3}(y''_n + y''_{n-1} + y''_{n-2}) + \frac{16}{240}h^6 y^{(6)}(\theta)$$

or

$$y_{n+1} = 2y_{n-1} - y_{n-3} + 4h^2(y''_{n-1} + \frac{\Delta^2 y''_{n-2}}{3}) + \frac{16}{240}h^6 y^{(6)}(\theta)$$

The predictor minus corrector gives

$$-\frac{17}{240}h^6 y^{(6)}(\theta)$$

so that the error in the corrector is about

$$-\frac{1}{17}(p_{n+1} - c_{n+1})$$

while the error in the predictor is

$$+\frac{1}{17}(p_n - c_n)$$

These can be used to improve the accuracy if desired.

The characteristic roots of the predictor are 1, -1 , i , $-i$ and give no trouble.

PROBLEMS

24.3.1 Derive the predictor equation.

24.3.2 Show that the predictor equation is a valid formula, by examining the G function.

24.4 LINEAR EQUATIONS

Linear differential equations occur very frequently, and in spite of the vast amount of theory associated with them it is sometimes necessary to solve one by numerical methods. The linear property makes them much easier to solve than the general equation. As an example, we consider the equation

$$y'' = f(x)y + g(x)$$

which is a special case of those in Sec. 24.3 (since y' does not occur). Using the corrector in the second form, we have

$$\Delta^2 y_{n-1} - \frac{h^2}{12} \Delta^2 y''_{n-1} = h^2 y''_n - \frac{h^6 y^{(6)}}{240}$$

Substituting the value for each appearance of y'' , we have

$$\Delta^2 y_{n-1} - \frac{h^2}{12} \Delta^2 [f(x_{n-1})y_{n-1} + g(x_{n-1})] = h^2 [f(x_n)y_n + g(x_n)]$$

$$\Delta^2 \left\{ \left[1 - \frac{h^2}{12} f(x_{n-1}) \right] y_{n-1} \right\} = h^2 f(x_n)y_n + h^2 \left(g_n + \frac{1}{12} \Delta^2 g_{n-1} \right)$$

We now write

$$\left[1 - \frac{h^2}{12} f(x_n) \right] y_n = Y_n$$

so that

$$\Delta^2 Y_{n-1} = h^2 [f(x_n)y_n + g_n + \frac{1}{12} \Delta^2 g_{n-1}]$$

Thus we solve

$$Y_{n+1} = 2Y_n - Y_{n-1} + h^2 [f(x_n)y_n + g_n + \frac{1}{12} \Delta^2 g_{n-1}]$$

where

$$y_n = \frac{Y_n}{1 - (h^2/12)f(x_n)}$$

This method is often attributed to Numerov [19] and is quite useful and efficient.

It should be evident that similar rearrangements of various methods to eliminate the predictor can be made when the equation is linear, and we need not give all the special cases explicitly.

PROBLEMS

24.4.1 Compute the solution of

$$y'' + xy = 0 \quad y(0) = 1 \quad y'(0) = 0$$

for $0 \leq x \leq 1$ and $h = 2/10$, using Numerov's method.

24.4.2 Show that no predictor is required for the linear equation

$$y' = P(x)y + Q(x)$$

24.5 A METHOD WHICH USES y , y' , AND y'' VALUES

We have pointed out several times that once a function has been computed, it is often quite economical of machine time to compute the derivative of the function. Thus, if we have

$$y' = f(x, y)$$

then

$$y'' = \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial x}$$

Problem 21.8.1 includes the formula ($b_{-1} = 1/2$)

$$y_{n+1} = y_n + \frac{h}{2}(y'_{n+1} + y'_n) + \frac{h^2}{12}(-y''_{n+1} + y''_n) + \frac{h^5}{720}y^{(5)}(\theta)$$

This appears to be an excellent choice for a corrector formula. Since only the y_n term occurs, there are no extraneous roots of the characteristic equation to produce instability. The truncation error is clearly very small, and roundoff is also under control.

We have, therefore, to find a predictor of the general form

$$y_{n+1} = A_0 y_n + A_1 y_{n-1} + h(B_0 y'_n + B_1 y'_{n-1}) + h^2(C_0 y''_n + C_1 y''_{n-1})$$

The usual process of making it exact for 1, x , x^2 , x^3 , x^4 produces the results given in Table 24.5. The choice $A_1 = 1$ has particularly attractive coefficients, although $A_1 = 0$ has a slightly smaller error.

PROBLEMS

24.5.1 Develop the details of a predictor-corrector method based on the $A_1 = 1$ case, including modification and final-value equations.

24.5.2 Examine the influence function for the case of $A_1 = 1$.

24.5.3 Derive

$$\begin{aligned} y_{n+1} = y_n + \frac{h}{240}(101y'_{n+1} + 128y'_n + 11y'_{n-1}) \\ + \frac{h^2}{240}(-13y''_{n+1} + 40y''_n + 3y''_{n-1}) \\ + \frac{8h^7 y^{(7)}(\theta)}{7!15} \end{aligned}$$

24.5.4 Derive, as in Prob. 24.5.3,

$$\begin{aligned} y_{n+1} = y_{n-1} + \frac{h}{15}(7y'_{n+1} + 16y'_n + 7y'_{n-1}) \\ + \frac{h^2}{15}(-y''_{n+1} + y''_{n-1}) + \frac{16h^7 y^{(7)}(\theta)}{7!15} \end{aligned}$$

24.5.5 Find a predictor to go with Probs. 24.5.3 and 24.5.4.

Table 24.5

	$A_1 = 0$	$A_1 = \frac{1}{2}$	$A_1 = 1$
$A_0 = 1 - A_1$	1	$\frac{1}{2}$	0
$A_1 = A_1$	0	$\frac{1}{2}$	1
$B_0 = (-1 + A_1)/2$	$-\frac{1}{2}$	$-\frac{1}{4}$	0
$B_1 = (3 + A_1)/2$	$\frac{3}{2}$	$\frac{7}{4}$	2
$C_0 = (17 - A_1)/12$	$\frac{17}{12}$	$\frac{31}{24}$	$\frac{5}{6}$
$C_1 = (7 + A_1)/12$	$\frac{7}{12}$	$\frac{13}{24}$	$\frac{2}{3}$
$E_5 = (31 + A_1)/6$	$\frac{31}{6}$	$\frac{25}{6}$	$\frac{16}{3}$

24.6 WHEN THE SOLUTION IS NOT EASILY APPROXIMATED BY A POLYNOMIAL

In general, we have been assuming that the functions which we are considering can be approximated by a polynomial, if we use reasonably sized steps. The main exception was in integration where we noted that the integral could be written in the form

$$\int_b^a f(x) dx = \int_b^a K(x)g(x) dx$$

and if the moments

$$m_k = \int_a^b K(x)x^k dx$$

are known and if $g(x)$ is easily approximated by a polynomial, then the integral can be solved readily. Thus we approximate only part of the integrand by a polynomial.

Somewhat similar situations arise in the field of differential equations; frequently the solution we are looking for cannot be closely approximated by a polynomial of reasonable degree, with steps of reasonable size. We then look to see what part of the problem we can approximate by a polynomial and try to take advantage of that.

As a simple example, suppose that we have

$$ay'' + by' + cy = f(x) \quad y(x_0) = y_0 \quad y'(x_0) = y'_0$$

where a , b , and c are constants and $f(x)$ is easily approximated by, say,

$$f(x) = p_i + q_i(x - x_{i-1}) \quad \text{for } x_{i-1} \leq x \leq x_i$$

Thus we are approximating $f(x)$ by a sequence of straight-line segments. It is easy to compute a particular solution of

$$y(x) = F_i(x) = P_i + Q_i(x - x_{i-1})$$

by direct substitution. The solution of the homogeneous differential equation leads to the characteristic equation

$$am^2 + bm + c = 0$$

with roots

$$m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Hence the solution of the differential equation in the i th interval ($x_{i-1} \leq x \leq x_i$) starts at

$$y(x_{i-1}) = y_{i-1}$$

$$y'(x_{i-1}) = y'_{i-1}$$

and is given by

$$y(x) = C_1 e^{m_1(x-x_{i-1})} + C_2 e^{m_2(x-x_{i-1})} + F_i(x)$$

To determine C_1 and C_2 , we use the initial conditions (of the interval)

$$y_{i-1} = C_1 + C_2 + F_i(x_{i-1})$$

$$y'_{i-1} = m_1 C_1 + m_2 C_2 + F'_i(x_{i-1})$$

We then compute the initial conditions for the next interval

$$y_{i-1}(x_i) = y_i$$

$$y'_{i-1}(x_i) = y'_i$$

and are ready for the next step.

If one or both of the values of m_i are large and negative, then the solution is not readily approximated by a polynomial, and by writing our solution in the form of a polynomial *plus* an exponential part, we have evaded the trouble. The approach used in no way depends on approximating $f(x)$ by linear polynomials, and generally speaking higher-order polynomials would be more economical of machine time. The approach also does not depend on the second-order equation; any order could have been used.

24.7 CONSERVATION LAWS

It often happens that the system of differential equations to be solved describes some quantities that obey conservation laws. For example, the total number of particles, molecules, or atoms is fixed (conserved). Such systems of equations have a form like

$$y'_i = f_i(t, y_1, y_2, \dots, y_n) \quad i = 1, 2, \dots, n$$

and if all the equations are added (sometimes suitable multiplicative constants¹ need to be supplied before adding), then we get the equation

$$\sum_{i=1}^n y'_i = 0$$

which implies

$$\sum_{i=1}^n y_i = C$$

¹ If variables are used as multipliers, for example to get the total energy, then the argument does not apply.

How will predictor-corrector methods treat the conservation? If we suppose that all the steps have obeyed the conservation up to the present step, then we find that the predicted values will also show the conservation (within roundoff) *provided* the right-hand sides of the equations have been arranged so that the large roundoffs that occur are carefully matched in the corresponding equations. And so it goes, step by step, with care in the evaluation of the right-hand sides so that any large roundoff in the right-hand side of one equation is compensated for by the corresponding equation; the system will, within roundoff, obey the conservation law.

This has an important consequence; often the user of the results will *assume* that if the conservation holds, then the results must be accurate. But we have just seen that the lack of conservation is a result of roundoff and *does not* involve the truncation error, and so the solutions may be very much in error—it is in the nature of predictor-corrector methods that the solution tends to obey the conservation.

24.8 STIFF EQUATIONS

We now examine another class of problems in which we do not approximate the final answer by a polynomial but only some part of it.

Large negative characteristic roots are typical of what are sometimes called “stiff equations.” The simplest form of a stiff equation is

$$y' = f(x, y) \quad \frac{\partial f}{\partial y} = A \ll 0$$

Such equations often arise in control systems where a rapid following of the signal is desired. They also occur in problems involving space charge as well as in flame chemistry.

One approach to the equation is based on the situation when $\partial f/\partial y$, though large and negative, is slowly changing. We write (for $A = \text{constant}$ approximately equal to $\partial f/\partial y$)

$$\begin{aligned} y' &= f(x, y) + Ay - Ay \\ y' - Ay &= f(x, y) - Ay \equiv F(x, y) \end{aligned}$$

where, of course,

$$\frac{\partial F}{\partial y} = \frac{\partial f}{\partial y} - A \approx 0$$

Solving the equation with the usual integrating factor e^{-Ax} , we have

$$y(x) = C_1 e^{A(x-x_0)} + e^{A(x-x_0)} \int_{x_0}^x e^{-A(\theta-x_0)} F[\theta, y(\theta)] d\theta$$

We now face the problem of deriving formulas for integrals of the form

$$I(h) = \int_0^h e^{-A\theta} g(\theta) d\theta$$

which is easy to do, following the methods developed so far. The predictor cannot use the value $g(h)$ but can use $g(0), g(-h), \dots$, as well as $I(0), I(-h), \dots$, while the corrector can use the predicted value of $g(h)$. We leave the details to the reader, but note that we are supposing that

$$f(x, y) = f(x, y) - Ay \quad [y = y(x)]$$

can be approximated by a polynomial using a fairly wide-spaced set of samples, and in many cases of stiff equations this is not so.

In the first example above we *added* the nonapproximatable part; in the second it was as a *multiplicative factor*. In both cases we abandoned the direct approximation of the solution by polynomials in favor of approximating some part of the solution.

Stiff equations are the topic of many papers these days, but it seems that we still do not understand the basic problem very well. Many of the proposed methods which damp out the oscillation also have a side effect of producing gross distortions in the solution being computed.¹ Material in Chaps. 31 to 41 present a framework of ideas for what the author thinks is a better approach.

PROBLEMS

24.8.1 Suppose that $f(x)$ can be approximated by a quadratic polynomial in the interval $0 \leq x \leq k2\pi$, and we have the equation

$$y'' + y = f(x)$$

Develop the above theory for numerically integrating the equation in the interval. Show how to extend it step by step to $2k\pi \leq x \leq 4k\pi$, $4k\pi \leq x \leq 6k\pi$, etc.

24.9 PROBLEMS WITH WIDELY DIFFERENT TIME CONSTANTS

The nature of problems with different time constants is perhaps best understood in terms of a concrete example. Suppose that a new chemical plant has been designed. The chemical engineers have carefully examined the chemical aspects of

¹ For a discussion of this fault, see, for example, E. E. Zajac, Note on Overly-Stable Difference Approximations, *J. Math. Phys.*, vol. 43, no. 1, pp. 51–54, 1964. See also M. E. Fowler and R. M. Warten, A Numerical Technique for Ordinary Differential Equations with Widely Separated Eigenvalues, *IBM, J. Res. Develop.*, September, 1967, pp. 537–543.

the problem, while the control engineers have examined their part of the design. It is now necessary to fill the inevitable gap that occurs when two groups of experts work on the same project; will the combined system operate properly? For this we take the differential equations of the complete system, chemistry and control, and set out to solve them.

The first thing that we find is that the control circuits tend to have rapid time constants, meaning, to be specific, that typically they will respond in time intervals of perhaps milliseconds, while the chemical engineering time response of the flow of materials, temperature changes, etc., may take up to an hour. Thus each simulated situation, for example how the plant will start up, will have to run perhaps an hour or more. But there are 3.6×10^6 msec in the hour, and the number of steps needed to integrate the equations is unpleasant to contemplate. This makes part of the system of equations look "stiff."

It often happens in such situations, not only in chemical plants but even in solid-state physics, that the system of equations can be broken into two sets, one with the long time constants and one with the short time constants. The short-time-constant equations tend to see the quantities from the long-time-constant equations as if they were constants, and thus their changes can pretty well be ignored in the integration of the control equations. On the other side, the control equations respond so rapidly that the slow-time-constant equations see them as being very close to equilibrium. In this situation the control differential equations can have their derivatives equated to zero and treated as algebraic or transcendental equations with no derivatives. Thus the two sets can be solved separately.

But beware! The purpose of the simulation was to see if the two groups of engineers overlooked any cross-connections between their two systems. Thus a careful mathematical examination must be made to ensure that everything said about the two time constants is in fact true, that neither is close to the other for *any* set of the variables. The outside examination of this point is a valuable contribution to the whole design problem, and it can be carried out with mathematical symbols which tend to have no preconceived ideas of how the answer should come out.

24.10 TWO-POINT PROBLEMS

We have been solving initial-value problems so far, but frequently conditions on the solution are given at two points. For example, we may have

$$y'' = f(x, y) \quad y(0) = 0 \quad y(1) = 0$$

and want to know $y = y(x)$ for $0 \leq x \leq 1$.

This situation may be reduced to the previous case by a trial-and-error process. We start with

$$y'' = f(x, y) \quad y(0) = 0 \quad y'(0) = \lambda$$

and try to find a pair of λ values λ_1 and λ_2 such that, say,

$$\begin{array}{ll} \text{for } \lambda_1 & y(1) < 0 \\ \text{for } \lambda_2 & y(1) > 0 \end{array}$$

Thus [since for most practical problems $y(1)$ is a continuous function of λ] we can use the crude process of bisecting the known range at each trial and hence in 10 trials reduce the range

$$|\lambda_1 - \lambda_2|$$

by about $2^{10} \approx 1,000$. More effective ways of locating the zero of $y(1)$ may be found easily. (See Chap. 4.)

Instead of reducing the two-point problem to a series of initial-value problems, it is more efficient to attack the problem directly. An extensive theory has been developed [13], and we shall briefly discuss only one example to show some of the ideas involved.

Suppose that we have

$$y'' = f(x)y + g(x) \quad y(0) = A \quad y(1) = B$$

We first approximate y'' by

$$h^2 y'' = \Delta^2 y_{n-1} \quad (24.10.1)$$

where we divided the interval $0 \leq x \leq 1$ into N intervals of size $h = 1/N$. Thus we have

$$y_{n+1} - 2y_n + y_{n-1} = h^2 [f(x_n)y_n + g(x_n)]$$

or

$$y_n = \frac{y_{n-1} + y_{n+1} - h^2 g(x_n)}{2 + h^2 f(x_n)} \quad n = 1, 2, \dots, N-1 \quad (24.10.2)$$

We have $N-1$ linear equations in $N-1$ unknowns, which may be solved in many ways.

As an example, let us consider a specific case of the problem. Set $A = B = 0$, $f(x) = 1$, $g(x) = x$, and $N = 4$. We have

$$y'' = y + x \quad y(0) = 0 \quad y(1) = 0$$

(which has the known solution $y = \sinh x / \sinh 1 - x$ for comparison). The difference equations are

$$\begin{aligned} y_2 - 2y_1 &= \frac{1}{16}(y_1 + \frac{1}{4}) \\ y_3 - 2y_2 + y_1 &= \frac{1}{16}(y_2 + \frac{2}{4}) \\ -2y_3 + y_2 &= \frac{1}{16}(y_3 + \frac{3}{4}) \end{aligned}$$

or

$$\begin{aligned} 16y_2 - 33y_1 &= \frac{1}{4} \\ 16y_3 - 33y_2 + 16y_1 &= \frac{2}{4} \\ -33y_3 + 16y_2 &= \frac{3}{4} \end{aligned}$$

One method of solving these particular equations is to multiply them by 16, 33, and 16, respectively, and add

$$(16^2 - 33^2 + 16^2)y_2 = \frac{1}{4}(16 + 2 \cdot 33 + 3 \cdot 16)$$

From y_2 it is easy to find y_1 and y_3 , if we use the first and third equations in turn.

Suppose that this is not as accurate as we wish; what shall we do next? We could set $N = 8$ and using the computed solution for $N = 4$, guess at a solution for $N = 8$. Then, using the guessed solution in the right-hand side of the eight equations corresponding to Eq. (24.10.2), we could compute improved values and repeat the process until no change occurs. This process can be extended as far as we wish.

Alternatively, and probably preferably, we could use a more accurate formula in place of Eq. (24.10.1). We use the S_7 matrix of Sec. 15.6 to find such a formula. The moments of $y''(0)$ are $(0, 0, 2, 0, 0, 0, 0)$ and should be written as a column vector. This produces the column vector corresponding to twice the third column of S_7 ; that is,

$$\frac{1}{360} \begin{pmatrix} 4 \\ -54 \\ 540 \\ -980 \\ 540 \\ -54 \\ 4 \end{pmatrix} = \frac{1}{180} \begin{pmatrix} 2 \\ -27 \\ 270 \\ -490 \\ 270 \\ -27 \\ 2 \end{pmatrix}$$

which is

$$\begin{aligned} h^2 y_n'' &= \frac{1}{180} (2y_{n+3} - 27y_{n+2} + 270y_{n+1} - 490y_n + 270y_{n-1} - 27y_{n-2} + 2y_{n-3}) \\ &= \Delta^2 y_{n-1} - \frac{1}{12} \Delta^4 y_{n-2} + \frac{1}{90} \Delta^6 y_{n-3} \end{aligned}$$

Using only the first two terms, since $\frac{1}{90} \Delta^6 y_{n-3}$ is likely to be small, we compute

$$-1/12 \Delta^4 y_{n-1}$$

To do this, we need some values outside the range $0 \leq x \leq 1$; in particular we need $y(-1/4)$ and $y(5/4)$. These can be found from the obvious equations

$$y_{-1} - 2y_0 + y_1 = h^2(y_0 + 0)$$

Alternatively, we note that we really need only $\Delta^2 y_{-1}$ and $\Delta^2 y_3$, which can be found from the values of y_0 and y_4 to be

$$\Delta^2 y_{-1} = 0 \quad \Delta^2 y_4 = h^2$$

Using the table of values,

	Δ	Δ^2	Δ^3	Δ^4
y_{-1}	_____			
y_0	_____	_____		
y_1	_____	_____	_____	_____
y_2	_____	_____	_____	_____
y_3	_____	_____	_____	_____
y_4	_____	_____		
y_5				

we can compute the values of $\Delta^4 y_i$, namely $\Delta^4 y_{-1}$, $\Delta^4 y_0$, and $\Delta^4 y_1$, required to improve the result.

These values of $\Delta^4 y_i$, which were neglected in the first computation, we now put in as *correction terms* on the right-hand side:

$$\Delta^2 y_{n-1} = h^2(y_n + x_n) + \frac{1}{12} \Delta^4 y_{n-3}$$

We solve the problem again, *without* changing the $\Delta^4 y_i$ terms. If the $\Delta^4 y_i$ values of the new solution are changed by any significant amount from the old ones, then we repeat the process *again*. This method is called the “difference-correction” method by Fox [13] who uses it with great effectiveness.

As already noted, we are not attempting to develop an elaborate theory for two-point problems but rather are content to give only a slight idea of what can be done with such problems.

PROBLEMS

24.10.1 Carry out the computation of the example in Sec. 24.10, using the difference-correction method, and compare the result with the correct answer.

25.1 INTRODUCTION

In Sec. 14.2 we discussed the four basic questions:

- 1 What samples shall we use?
- 2 What class of functions shall we use?
- 3 What criterion shall we use?
- 4 Where shall we apply the criterion?

We have used both function and derivative values as samples (as well as using the position of the samples as parameters), but we have used only the class of polynomials and the criterion of exactly matching the samples. We now begin the examination of other criteria for picking the particular member of the class of functions being used. While we shall continue to concentrate on polynomials as the class of functions, more general classes will be mentioned from time to time.

In this chapter and in Chaps. 26 and 27, we examine the criterion of least squares, and in Chaps. 28 and 29, we examine the Chebyshev, or minimax, criterion. These are the two most widely used criteria excluding exactly matching at the sample points (within roundoff, of course), but others do arise now and then.

Having seen three different examples used in the book, hopefully the reader will be able to handle others when they arise.

The least-squares criterion is used in situations where there is much more data available than parameters so that exact matching is out of the question. Polynomials are most commonly used in least-squares matching, though any linear family will work about as well in practice.

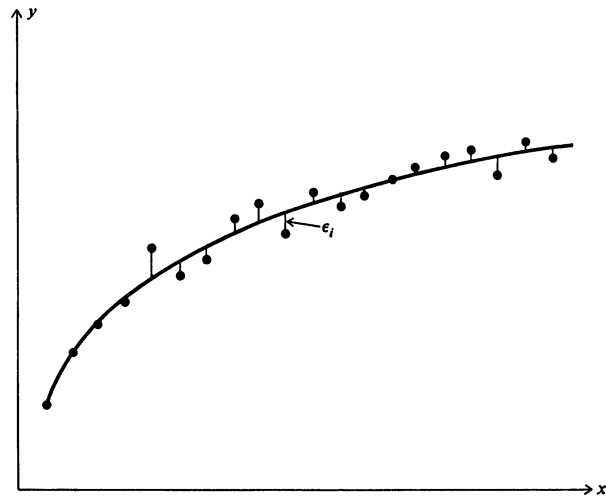


FIGURE 25.1.1

The least-squares criterion is widely used, and often believed to be the “right one” to use. There is a saying that mathematicians believe that it is a physical principle while physicists believe that it is a mathematical principle. Either we can assume the principle for selecting the particular member of the family of polynomials of a given degree, or we can assume some other principles and deduce that of least squares—something must be assumed in any case.

Least squares is very often regarded as *smoothing the data*, or, if you wish, *removing the noise*. The least-squares polynomial is supposed to give the correct values (Fig. 25.1.1). In order to understand when to use the principle and when to avoid it, we shall try to give a feeling for what it does by connecting it with other known results. Just what this kind of smoothing does will be looked at again in Secs. 27.7 and 35.6.

25.2 THE PRINCIPLE OF LEAST SQUARES

Suppose we try to measure some quantity x and make M measurements x_i (see Fig. 25.2.1). We do not see x but only the measurements x_i with errors of measurement ε_i ; that is, we see

$$x_i = x + \varepsilon_i \quad i = 1, 2, \dots, M$$

We will regard the residuals ε_i as “noise” and call x the true value, (whatever that may mean in a situation in which it cannot be measured directly!).

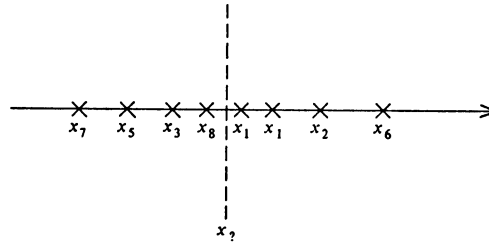


FIGURE 25.2.1

The principle of least squares states that the best estimate $\hat{x}$ of x is that number which minimizes the sum of the squares of the deviations of the data from their estimate,¹

$$f(\hat{x}) = \sum_{i=1}^M \varepsilon_i^2 = \sum_{i=1}^M (x_i - \hat{x})^2$$

In the final analysis, the usefulness of this principle rests on how useful the results turn out to be in practice and how easy it is to use, not on the fanciness of the mathematical derivations.

This principle is equivalent to the assumption that the average

$$x_a = \frac{1}{M} \sum_{i=1}^M x_i$$

is the best estimate. To prove this equivalence, we first show that the least-squares principle leads to the average. We regard

$$f(\hat{x}) = \sum_{i=1}^M (x_i - \hat{x})^2$$

¹ The Gauss-Markov theorem states that if $\text{Av}\{\varepsilon_i\} = 0$ and $\text{cov}\{\varepsilon_i, \varepsilon_j\} = 0$, then of all unbiased linear estimates this estimate has the least variance. See Mood and Graybill [42].

as a function of $\hat{x}$ to be minimized. Applying the usual calculus rule, “differentiate and set equal to zero,” we get

$$\begin{aligned}\frac{df}{d\hat{x}} &= -2 \sum_{i=1}^M (x_i - \hat{x}) = 0 \\ \sum_{i=1}^M x_i - \sum_{i=1}^M \hat{x} &= 0 \\ \sum_{i=1}^M x_i - M\hat{x} &= 0\end{aligned}$$

therefore

$$\hat{x} = \frac{1}{M} \sum_{i=1}^M x_i = x_a$$

Thus $x_a = \hat{x}$ minimizes the sum of the squares of the residuals. We also note that

$$\frac{d^2f}{d\hat{x}^2} = 2M > 0$$

and hence we have a minimum.

We now show the converse, namely that if we pick the average x_a as the best choice, then this minimizes the sum of the squares. We set

$$\begin{aligned}f(x_a) &= \sum_{i=1}^M (x_i - x_a)^2 \\ &= \sum x_i^2 - 2x_a \sum x_i + \sum x_a^2 \\ &= \sum x_i^2 - 2x_a Mx_a + Mx_a^2 \\ &= \sum x_i^2 - Mx_a^2\end{aligned}$$

Now if we were to choose any other value, say x_b , then

$$\begin{aligned}f(x_b) &= \sum (x_i - x_b)^2 \\ &= \sum x_i^2 - 2x_b Mx_a + Mx_b^2\end{aligned}$$

Subtracting the first result from the second, we get

$$\begin{aligned}f(x_b) - f(x_a) &= M[x_a^2 - 2x_a x_b + x_b^2] \\ &= M(x_a - x_b)^2 \geq 0\end{aligned}$$

Thus

$$f(x_b) \geq f(x_a)$$

and they are equal only if $x_b = x_a$; that is, x_a minimizes the sum of the squares of the residuals.

We have now proved that the principle of the least squares and the choice of the average as the best value are equivalent.

The choice of the average as the best value is very common practice in many situations, but clearly is inappropriate in some. For example, the average income in a country is likely to be misleading because of the skewness of the distribution of incomes. Likewise, if the probability distribution of the errors (residuals) is believed to be highly skewed, then it may be foolish to use the principle of least squares.

Probably the major fault with least squares is that a single very wrong measurement will greatly distort the results because in the squaring process large residuals play the dominant part — one gross error 10 times larger than most of the others will have the same effect in the sum of the squares as will 100 of the others. Great care should be exercised before blindly applying any result to data (as is so often done); at least look at the residuals, either by eye or by some suitable program, to see if one or possibly a few measurements are wildly off.

PROBLEMS

25.2.1 Find the least-squares estimate of the numbers 2, 3, 2, 1, 2, 3, both ways.

Ans. $x = 13/6$

25.2.2 Find the least-squares estimate of $-5, 5, 4, -3, 7, 1, 0, 0, 4, -6$

25.3 OTHER CHOICES BESIDES LEAST SQUARES

Suppose that instead of choosing to minimize the sum of the squares of the deviations ε_i we choose x_m to minimize the sum of the absolute values

$$\sum_{i=1}^M |\varepsilon_i| = \sum_{i=1}^M |x_i - x_m| = f(x)$$

This leads to the choice of the *median* (middle) value of the x_i (if there is an even number of observations M , then we choose the average of the two middle ones). The proof is straightforward. Suppose that there is an odd number $M = 2k + 1$ of observations x_i . Consider using the middle one, x_m , as the median. Then any upward shift of this value x would increase the k terms $|x_i - x|$ that have x_i below x_m and would decrease the k values with x_i above x_m , each by the same amount, but the shift would *also* increase the term $|x_m - x|$, thus increasing the sum of all the deviations.

In place of the minimizing of the sum of the squares, another choice is to minimize the maximum deviation. This leads to

$$\frac{x_{\min} + x_{\max}}{2} = x_{\text{midrange}}$$

which is the usual *midrange* estimate of the best value. This is the Chebyshev criterion in slight disguise and will be treated in Chaps. 28 and 29.

25.4 THE NORMAL LAW OF ERRORS

There appears to be a widespread belief that the principle of least squares implies *the normal law of errors*, which states that the probability of an error ε_i in the interval $(x, x + \Delta x)$ is given by

$$\frac{k}{\sqrt{\pi}} e^{-k^2 x^2} \Delta x$$

This belief is false, and it is worth some discussion of the topic at this point to clarify matters. Another belief sometimes met is that the normal law is a law of nature. This situation has also been paraphrased by the remark, “mathematicians believe it to be a physical law, while physicists believe it to be a mathematical law.”

One highly seductive derivation of the normal law (due to Herschel¹) goes as follows: Consider dropping a dart from a height while aiming at some point 0 on the (horizontal) floor (see Fig. 25.4.1). We now *assume* (1) that the errors do not depend on the coordinate system but only on the distance from 0, and (2) that larger errors are less likely than smaller ones. These assumptions seem to be quite reasonable, since the coordinate system appears to be perfectly arbitrary.

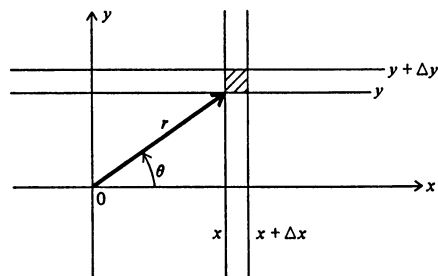


FIGURE 25.4.1

Let the probability of falling in the strip $x, x + \Delta x$ be given (approximately) by

$$f(x) \Delta x$$

while that of falling in the strip $y, y + \Delta y$ is

$$f(y) \Delta y$$

¹ Sir John F. W. Herschel (1792–1871).

Owing to the assumption that the errors in the two coordinates are independent, the probability that it falls in the common rectangle is then

$$f(x)f(y) \Delta x \Delta y$$

Let us now choose polar coordinates. Since the probability is assumed to be independent of direction, the probability that the dart falls in the element of area is given by

$$g(r) \Delta x \Delta y$$

The two probabilities are the same, and we have

$$g(r) = f(x)f(y)$$

The left-hand side does not depend on θ , and differentiating with respect to θ , we have

$$\frac{\partial g(r)}{\partial \theta} = 0 = f(x) \frac{\partial f(y)}{\partial \theta} + f(y) \frac{\partial f(x)}{\partial \theta} \quad (25.4.1)$$

Using the relations

$$x = r \cos \theta$$

$$y = r \sin \theta$$

we have

$$\begin{aligned} \frac{\partial f(x)}{\partial \theta} &= \frac{\partial f(x)}{\partial x} \frac{\partial x}{\partial \theta} = f'(x)(-y) \\ \frac{\partial f(y)}{\partial \theta} &= \frac{\partial f(y)}{\partial y} \frac{\partial y}{\partial \theta} = f'(y)(x) \end{aligned} \quad (25.4.2)$$

Combining Eqs. (25.4.2) and (25.4.1), we have

$$f(x)f'(y)(x) + f(y)f'(x)(-y) = 0$$

or

$$\frac{f'(x)}{xf(x)} = \frac{f'(y)}{yf(y)}$$

Since x and y are assumed to be independent, each side must be a constant, say K , and

$$\frac{f'(x)}{xf(x)} = K = \frac{f'(y)}{yf(y)}$$

or, what is the same thing,

$$\frac{df(x)}{f(x)} = K \cdot x \cdot dx \quad \text{and} \quad \frac{df(y)}{f(y)} = K \cdot y \cdot dy$$

Integrating the first of these, we get

$$\ln f(x) = \frac{Kx^2}{2} + C$$

or

$$f(x) = Ae^{Kx^2/2}$$

We assumed that large errors were less likely than small ones; hence K must be negative, say

$$K = -2k^2$$

and so we have

$$\begin{aligned} f(x) &= Ae^{-k^2x^2} \\ f(y) &= Ae^{-k^2y^2} \end{aligned}$$

Combining them, we get

$$g(r) = f(x)f(y) = A^2e^{-k^2(x^2+y^2)}$$

Now the dart must fall somewhere; hence, using polar coordinates, we find

$$\begin{aligned} \int_0^{2\pi} \int_0^\infty g(r)r \, dr \, d\theta &= 1 \\ A^2 \cdot 2\pi \int_0^\infty re^{-k^2r^2} \, dr &= 1 \\ A^2\pi \left. \frac{e^{-k^2r^2}}{-k^2} \right|_0^\infty &= \frac{A^2\pi}{k^2} = 1 \end{aligned}$$

and finally we get our result

$$f(x) = \frac{k}{\sqrt{\pi}} e^{-k^2x^2}$$

which is the normal (Gaussian) distribution function.

It should be noted that this derivation is only one way of obtaining the normal law. The central-limit theorem, which states, loosely, that the distribution of the sum of a large number of small errors becomes normal, provides another approach. In either case, it is not the derivation that matters as much as whether or not the model helps us understand experience.

The normal law has been found in practice to be a useful model in many applications. Deviations from it usually occur from having more values in the “tail” of the distribution when x is large than the model indicates there should be. The reason for this is that often there is a small effect which has a wide variability. In such cases, a mixture of two normal curves with different k values sometimes is useful. The theory of *quality control* is, in part, based on the observed excesses in the tails.

The *maximum-likelihood estimator* [42] of an unknown parameter maintains that we should take the value that maximizes the likelihood of observing it. If the sample is from some probability density $p(x; \theta)$, then the likelihood estimator is the product of the individual (independent) observations

$$L(\theta) = p(x_1; \theta)p(x_2; \theta) \cdots p(x_m; \theta)$$

In the case that the errors come from a normal distribution

$$\frac{k}{\sqrt{\pi}} e^{-k^2(x-\theta)^2}$$

then this leads to

$$L(\theta) = \frac{k^M}{\pi^{M/2}} e^{-k^2 \sum (x_i - \theta)^2}$$

When this is maximized it clearly is the least-squares estimate and leads to

$$\theta = \frac{1}{M} \sum_{i=1}^M x_i = \bar{x}$$

Thus least squares can be derived from the normal distribution via the (assumed) maximum-likelihood estimator.

PROBLEMS

25.4.1 Show that $\sigma^2 = 1/(2k)^2$, where σ is the variance of the distribution $f(x) = (k/\sqrt{\pi})e^{-k^2x^2}$.

25.5 THE LEAST-SQUARES STRAIGHT LINE

Before examining the general case of a polynomial, it is worth becoming familiar with the special case of fitting a straight line in the least-squares sense. We fit

$$y(x) = a + bx$$

to a set of data (x_i, y_i) ($i = 1, 2, 3, \dots, M$). We have two parameters a and b and want to minimize the function of two variables (a and b)

$$F(a, b) = \sum_{i=1}^M [y(x_i) - y_i]^2 = \sum_{i=1}^M \varepsilon_i^2$$

$y(x_i)$ being the computed value and y_i the measured value. This is the obvious generalization of the material on a single parameter that we discussed before. We have

$$F(a, b) = \sum_{i=1}^M (a + bx_i - y_i)^2$$

to minimize. Apply the calculus method

$$\frac{\partial F(a, b)}{\partial a} = 2 \sum (a + bx_i - y_i) = 0$$

$$\frac{\partial F(a, b)}{\partial b} = 2 \sum (a + bx_i - y_i)x_i = 0$$

Dropping the factor 2 and rewriting, we get

$$\begin{aligned} aM + b \sum x_i &= \sum y_i \\ a \sum x_i + b \sum x_i^2 &= \sum x_i y_i \end{aligned}$$

These equations are easily solved for the parameters a and b . Thus we have the least-squares straight line to the given data. As an easily followed example, consider the data (for $M = 5$). See Fig. 25.5.1.

x	y	x^2	xy
1	0	1	0
2	2	4	4
3	2	9	6
4	5	16	20
5	4	25	20
Sum	15	55	50

The equations for the parameters are:

$$\begin{aligned} 5a + 15b &= 13 & -3 \\ 15a + 55b &= 50 & 1 \end{aligned}$$

$$10b = 11$$

$$\begin{cases} b = 11/10 \\ a = -7/10 \end{cases}$$

The line is

$$y = \frac{1}{10}(-7 + 11x)$$

To plot the line we use

$$y(0) = -7/10$$

$$y(5) = 48/10$$

We have

x	$y(x)$	y_i	ε_i	ε_i^2
1	0.4	0	0.4	0.16
2	1.5	2	-0.5	0.25
3	2.6	2	0.6	0.36
4	3.7	5	-1.3	1.69
5	4.8	4	0.8	0.64
$\Sigma = 0$				3.10
check				

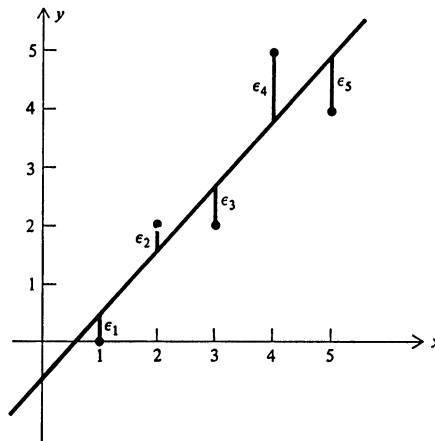


FIGURE 25.5.1

PROBLEMS

Fit a straight line to:

25.5.1	x	-3	-2	-1	0	1	2	3
	y	4	4	2	2	1	0	0

25.5.2	x	2.5	1.7	3.2	0.7	0.3
	y	1.7	1.4	2.5	0.6	0.1

25.5.3	x	1	2	3	4	5	6
	y	0	1	4	9	16	25

25.5.4 $y = e^x$ for 0, 2/10, 4/10, ..., 1.025.5.5 Prob. 25.5.4 for e^{-x} .**25.6 POLYNOMIAL CURVE FITTING**

Fitting a polynomial of degree N to some data is a direct generalization of the previous section. Given the M measurements (observations) (x_i, y_i) , we wish to fit a polynomial of degree N where N is much less than M , typically at least 10 times smaller for cubics or higher-degree polynomials. We again let the calculated data be $y(x_i)$ where

$$y(x) = a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N$$

and form the sum of the squares of the differences (residuals)

$$\sum_{i=1}^M [y(x_i) - y_i]^2 = F(a_0, a_1, a_2, \dots, a_N)$$

With $N + 1$ parameters we have the $N + 1$ partial derivatives to equate to zero

$$\frac{\partial F}{\partial a_j} = 2 \sum_{i=1}^M [(y(x_i) - y_i)x_i^j] = 0 \quad j = 0, 1, \dots, N$$

or

$$a_0 \sum x_i^j + a_1 \sum x_i^{j+1} + \dots + a_N \sum x_i^{j+N} = \sum x_i^j y_i$$

For ease in notation we set

$$\sum_{i=1}^M x_i^k = S_k \quad \sum_{i=1}^M x_i^k y_i = T_k$$

The equations then become

$$\sum_{n=0}^N a_n S_{n+j} = T_j \quad j = 0, 1, \dots, N$$

and are the *normal equations*.

The normal equations are $N + 1$ linear equations in $N + 1$ unknowns a_j , whose determinant we now show cannot be zero. We do this exactly as we did in Sec. 19.4. We consider the homogeneous equations

$$\sum_{n=0}^N a_n S_{n+j} = 0$$

and show that they have only the trivial solution $a_j = 0$ for all j , from which it follows that the determinant is not zero. Multiply the j th equation by a_j and sum for all j

$$\begin{aligned} \sum_{j=0}^N \sum_{n=0}^N a_j a_n S_{n+j} &= \sum_{j=0}^N \sum_{n=0}^N a_j a_n \sum_{i=1}^M x_i^j x_i^n \\ &= \sum_{i=1}^M \left(\sum_{j=0}^N a_j x_i^j \right) \left(\sum_{n=0}^N a_n x_i^n \right) \\ &= \sum_{i=1}^M y^2(x_i) = 0 \end{aligned}$$

This means that $y(x) \equiv 0$ and $a_j = 0$ for all j . We have therefore proved that the determinant is not zero and that the nonhomogeneous equations can be solved for the coefficients of the least-squares polynomial.

In principle the problem of finding the least-squares polynomial is solved; in practice it is not easy to solve the normal equations when N is very large, say greater than 10. To see why this is so, suppose that the x_i are more or less uniformly distributed in the interval $0 \leq x \leq 1$. Then

$$S_k = \sum_{j=1}^M x_j^k \approx M \int_0^1 x^k dx = \frac{M}{k+1}$$

The resulting determinant (suppressing the factors M) is the well-known Hilbert determinant

$$\left| \frac{1}{j+k+1} \right| \quad j, k = 0, 1, \dots, N$$

The Hilbert determinant¹ of order N has the value

$$H_N = \frac{N!(N+1)! \cdots (2N-1)!}{[1!2!3! \cdots (N-1)!]^3}$$

which approaches zero very rapidly and suggests what is true, that the system of normal equations will be difficult to solve (ill-conditioned, Sec. 7.8) when N is even moderate in size. The following table gives some values

n	H_n
1	1
2	8.3×10^{-2}
3	4.6×10^{-4}
4	1.7×10^{-7}
5	3.7×10^{-12}
6	5.4×10^{-18}
7	4.8×10^{-25}
8	2.7×10^{-33}
9	9.7×10^{-43}

The solution to this dilemma is given in the next chapter, and it is *not* to go to multiple precision to get an answer to a set of equations which in their very formation have significant errors.

As an example of polynomial curve fitting consider fitting a quadratic

$$y(x) = a + bx + cx^2$$

to the following data. The given data are in the two left-hand columns ($M = 7$) and are chosen to make the example easy to follow.

x	y	xy	x^2	x^2y	x^3	x^4
-3	4	-12	9	36	-27	81
-2	2	-4	4	8	-8	16
-1	3	-3	1	3	-1	1
0	0	0	0	0	0	0
1	-1	-1	1	-1	1	1
2	-2	-4	4	-8	8	16
+3	-5	-15	9	-45	27	81
Sums	0	1	-39	28	-7	196

¹ David Hilbert (1862–1943).

The normal equations are

$$7a + 0b + 28c = 1$$

$$0a + 28b + 0c = -39$$

$$28a + 0b + 196c = -7$$

It follows easily that $b = -39/28$. The other two equations can be written

$$\begin{cases} a + 4c = 1/7 \\ a + 7c = -1/4 \end{cases}$$

from which

$$a = 56/84$$

$$c = -11/84$$

and

$$y = \frac{1}{84} (56 - 117x - 11x^2)$$

which is shown in Fig. 25.6.1.

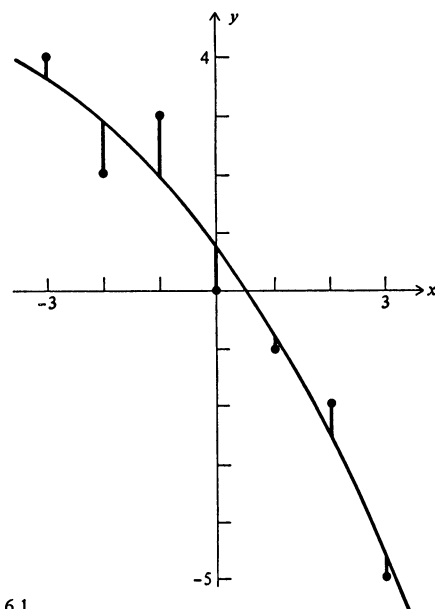


FIGURE 25.6.1

PROBLEMS

25.6.1 Fit a quadratic to

x	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
y	-3	-3	-2	1	2	2

25.6.2 Fit a quadratic to

x	0	1	2	3	4	5
y	4	7	10	13	16	19

25.6.3 Draw a flow diagram for doing least squares, assuming a library program for solving the normal equations. Find the residuals.

25.7 NONPOLYNOMIAL LEAST SQUARES
AND OTHER GENERALIZATIONS

If instead of fitting the data in terms of a linear combination of powers of x , suppose we are given a linearly independent set of functions $f_j(x)$ as the basis for the representation. We have

$$y(x) = a_0 f_0(x) + a_1 f_1(x) + \cdots + a_N f_N(x)$$

and the least-squares fit is given by

$$\sum_{i=1}^M \left[\sum_{j=0}^N a_j f_j(x_i) - y_i \right]^2 = \min$$

Differentiating this with respect to the parameters a_k (which still occur linearly) we are led to the system of equations

$$\sum_{i=1}^M \sum_{j=1}^N a_j f_j(x_i) f_k(x_i) = \sum_{i=1}^M y_i f_k(x_i)$$

Using the notation of Sec. 25.6, we set correspondingly

$$S_{j,k} = \sum_{i=1}^M f_j(x_i) f_k(x_i) \quad T_k = \sum_{i=1}^M y_i f_k(x_i)$$

and we have similar normal equations

$$\sum_{j=1}^N a_j S_{j,k} = T_k \quad k = 0, 1, \dots, N$$

The same proof that the determinant is nonzero will go through since the family of functions was assumed to be linearly independent over the set of points x_i .

If all the data are not equally reliable and it is desirable to weight the data

by nonnegative constants w_i , then again we will be led to the least-squares equation

$$\sum_{i=1}^M w_i [\sum a_j f_j(x_i) - y_i]^2 = \min$$

and the normal equations will include the weights w_i with nothing significantly different. The linear independence of the $f_j(x_i)$ means that the determinant will not be zero (by the same pattern of proof).

The case where the parameters do not occur linearly will be discussed in more detail in Sec. 27.5.

PROBLEMS

25.7.1 Carry out the derivation for weights $w_i \geq 0$ and prove that the normal equation can be solved.

25.8 A COMPARISON OF LEAST SQUARES AND POWER-SERIES EXPANSION

The power-series expansion of a function $y(x)$,

$$y(x) = \sum_{n=0}^{\infty} \frac{y^{(n)}(0)x^n}{n!}$$

when truncated to

$$y_N(x) = \sum_{n=0}^N \frac{y^{(n)}(0)x^n}{n!}$$

gives a very good approximation near $x = 0$, but as x increases, the approximation tends to get worse and worse. On the other hand, the least-squares fit tries to find a more or less uniform [depending on the weight function $w(x)$] approximation *in an interval*. And this is one of the fundamental differences: The power series fits at a point whereas the least squares fits in an interval.

When it comes to numerical computation, it is not generally possible to estimate accurately the derivatives at a point from the samples scattered in an interval; the exact-matching interpolating polynomial is frequently used in place of the truncated power series, or sometimes a least-squares polynomial.

25.9 CONCLUDING REMARKS ON LEAST SQUARES

Having been shown how to produce a least-squares polynomial, the reader probably expects the text to go on to the problems of integrals and differential equations. While this could be done, the facts are that seldom is a least-squares approximation used as a basis for integration. Integration itself is a "smoothing operation." When there is a "noisy" term in a differential equation, it is customary to smooth it and then use an exact-matching polynomial for integrating the equation.

Another idea which may occur to the reader, but which has not been adequately explored, is the following: In the exact-matching process, we have used $E_0, E_1, \dots, E_{m-1} = 0$ and let the rest, $E_m, E_{m+1}, \dots$, fall where they may. Suppose that we were to try to minimize

$$m(a_0, a_1, \dots) = \sum_{k=0}^{\infty} a_k E_k^2$$

The exact matching used $a_m = a_{m+1} = \dots = 0$, but we could taper the a_k and thus allow for an improved fit for some of the higher powers at the cost of some exactness for the lower powers.

This can be viewed as a proposal to fit in error space rather than in the function space. An adequate theory for the error term is not known at present.

An example of this proposal to sacrifice a small amount of exactness in fitting at some value for a large gain at some higher value of k is given by the idea of throwback, discussed briefly in Sec. 18.8 as applied to Everett's interpolation formula. In the particular case, we gave up a small amount of accuracy in x^2 (modified Δ^2) for a large gain in fitting x^3 and x^4 . (See also Sec. 29.9.)

26

ORTHOGONAL FUNCTIONS

26.1 INTRODUCTION

The idea of orthogonal functions plays a central role in the development of many branches of mathematics. We shall confine our attention to the relationship of orthogonality to linear independence and least squares, and the role that it plays in some parts of computing. Thus we treat only a few aspects of the topic in this chapter.

Two vectors x and y are said to be *orthogonal* if the components x_i and y_i satisfy the equation

$$\sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + x_3 y_3 + \cdots + x_n y_n = 0$$

In both two and three dimensions this means that the two vectors are perpendicular, and since the algebraic relationship plays the same role in higher dimensions as in two and three dimensions, it is customary to say that they are perpendicular to each other in n -dimensional space.

Now let us imagine that the number of dimensions increases to infinity in

such a way that the vectors are replaced in the limit by continuous functions, and that the sum approaches an integral. We then have

$$\int_a^b x(t)y(t) dt = 0$$

Thus we have the definition that two functions are orthogonal if the product of the two functions integrated over the range is zero. We have tacitly excluded the vector and the function that are identically zero, since they are perpendicular to all functions including themselves. Notice that if we are going to maintain the geometric language, then we have an infinitely nondenumerable dimensional space of functions.

Sometimes it is convenient to include a weight factor $w(t) \geq 0$, and the definition of orthogonality becomes

$$\int_a^b w(t)x(t)y(t) dt = 0$$

A set of functions $f_i(t)$ is said to be *orthogonal* [with respect to the weight factor $w(t)$ and over the interval $a \leq x \leq b$] if

$$\int_a^b w(t)f_i(t)f_j(t) dt = \begin{cases} 0 & i \neq j \\ \lambda_i > 0 & i = j \end{cases}$$

If

$$\int_a^b w(t)f_i^2(t) dt = 1$$

we say that the system is *orthonormal*. It is easy to make an orthogonal set orthonormal if it is not. To do this, if

$$\int_a^b w(t)f_i^2(t) dt = \lambda_i > 0$$

then the set $f_i(t)/\sqrt{\lambda_i}$ is orthonormal.

We have defined the idea of orthogonality over a continuous interval. There are analogous definitions over a finite set of points, although in the latter case there can be at most as many mutually orthogonal functions as there are points in the set. It is convenient to use the continuous version most of the time and only occasionally mention the discrete case.

26.2 SOME EXAMPLES OF ORTHOGONAL SYSTEMS OF FUNCTIONS

Perhaps the best-known set of orthogonal functions, and certainly the one that has given rise to the most new mathematics (including Lebesgue integration), is the family

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \dots$$

We first prove that they are orthogonal over the interval $0 \leq x \leq 2\pi$. We need to show that

$$\begin{aligned}\int_0^{2\pi} \cos mx \cos nx \, dx &= \begin{cases} 2\pi & m = n = 0 \\ \pi & m = n \neq 0 \\ 0 & m \neq n \end{cases} \\ \int_0^{2\pi} \cos mx \sin nx \, dx &= 0 \\ \int_0^{2\pi} \sin mx \sin nx \, dx &= \begin{cases} \pi & m = n \neq 0 \\ 0 & m \neq n \end{cases}\end{aligned}$$

In the first case, using the standard trigonometric identity, we get

$$\begin{aligned}\int_0^{2\pi} \cos mx \cos nx \, dx &= \frac{1}{2} \int_0^{2\pi} [\cos(m+n)x + \cos(m-n)x] \, dx \\ &= \frac{1}{2} \left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n} \right] \Big|_0^{2\pi} = 0 \quad m - n \neq 0\end{aligned}$$

If $m = n \neq 0$, then we have

$$\int_0^{2\pi} \cos mx \cos nx \, dx = \frac{1}{2} \int_0^{2\pi} [\cos(m+n)x + 1] \, dx = \pi$$

If $m = n = 0$, then we have

$$\int_0^{2\pi} \cos mx \cos nx \, dx = \int_0^{2\pi} 1 \, dx = 2\pi$$

The other relations can be proved similarly.

We can see a simple use for orthogonality in the formal expansion of a function; for example, suppose for some function $F(x)$ ($0 \leq x \leq 2\pi$)

$$\begin{aligned}F(x) &= \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots \\ &= \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)\end{aligned}$$

Multiply both sides by $\cos mx$ and integrate ($0 \leq x \leq 2\pi$)

$$\int_0^{2\pi} F(x) \cos mx \, dx = \pi a_m \quad (m = 0, 1, \dots)$$

(all other terms vanish due to the orthogonality).

Using $\sin mx$ in place of $\cos mx$, we get

$$\int_0^{2\pi} F(x) \sin mx \, dx = \pi b_m \quad m = 1, 2, \dots$$

Thus using orthogonality, we can formally find the individual coefficients of the expansion. The use of $a_0/2$ is for convenience.

The a_m and b_m computed this way are called *the Fourier¹ coefficients*, even in the case of a general system of orthogonal functions

$$\int_a^b w(x)f_i(x)f_j(x) dx = \begin{cases} 0 & i \neq j \\ \lambda_j & i = j \end{cases}$$

If

$$F(x) = \sum_{i=0}^{\infty} a_i f_i(x)$$

then

$$a_j = \frac{1}{\lambda_j} \int_a^b w(x)F(x)f_j(x) dx$$

are the Fourier coefficients.

As another example of orthogonal functions, consider the ultraspherical polynomials defined by the differential equation

$$(1-x^2)y''_n(x) - 2(\alpha+1)xy'_n(x) + n(n+2\alpha+1)y_n(x) = 0 \quad \alpha > -1$$

If we multiply this equation by $(1-x^2)^\alpha y_m(x)$, ($m \neq n$) and integrate ($-1 \leq x \leq 1$) we get

$$\begin{aligned} \int_{-1}^1 y_m(x) \frac{d}{dx} [(1-x^2)^{\alpha+1} y'_n(x)] dx + n(n+2\alpha+1) \\ \times \int_{-1}^1 (1-x^2)^\alpha y_m(x)y_n(x) dx = 0 \end{aligned}$$

Integrate the first integral by parts

$$\begin{aligned} y_m(x)(1-x^2)^{\alpha+1} y'_n(x) \Big|_{-1}^1 - \int_{-1}^1 (1-x^2)^{\alpha+1} y'_m(x)y'_n(x) dx \\ + n(n+2\alpha+1) \int_{-1}^1 (1-x^2)^\alpha y_m(x)y_n(x) dx = 0 \end{aligned}$$

The integrated part vanishes at both limits. We next write the same equation with m and n interchanged and subtract the two equations to get

$$[m(m+2\alpha+1) - n(n+2\alpha+1)] \int_{-1}^1 (1-x^2)^\alpha y_m(x)y_n(x) dx = 0$$

For $m \neq n$ the term inside the square bracket is not zero; hence for all $\alpha > -1$ the ultraspherical polynomials are orthogonal over the interval $-1 \leq x \leq 1$ with the weight factor $(1-x^2)^\alpha$. Thus we have shown that the polynomial solutions of the differential equation for different n are orthogonal.

This family, depending on α , of orthogonal polynomial systems includes many special cases of importance which we will later study, $\alpha = -\frac{1}{2}, 0, \frac{1}{2}$, and 1.

¹ Jean Baptiste Joseph Fourier (1768–1830).

PROBLEMS

- 26.2.1 Complete the proof of the orthogonality of the trigonometric functions.
 26.2.2 Prove directly the orthogonality of the set of trigonometric functions over $-\pi \leq x \leq \pi$.
 26.2.3 For the set of functions $1, \cos x, \cos 2x, \dots$ show that they are orthogonal ($0 \leq x \leq \pi$).
 26.2.4 For the set of functions $\sin x, \sin 2x, \dots$ show that they are orthogonal ($-\pi/2 \leq x \leq \pi/2$).
 26.2.5 Show that for the case of $\alpha = -\frac{1}{2}$ in the ultraspherical polynomials the substitution $x = \cos \theta$ produces a familiar situation.
 26.2.6 Normalize the Fourier-series terms.

26.3 LINEAR INDEPENDENCE AND ORTHOGONALITY

The purpose of this section is to show that the ideas of linear independence and orthogonality are closely related. First, we prove that *a set of orthogonal functions* $f_i(x)$ is linearly independent over the interval. Suppose that there is a linear relation among the $f_i(x)$ with nonzero coefficients, that is, that there exists a relation

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_N f_N(x) \equiv 0$$

with some $c_j \neq 0$. Multiply this relation through by $w(x)f_j(x)$ and integrate over the interval. We get

$$\begin{aligned} c_1 \int_a^b w(x) f_1(x) f_j(x) dx + c_2 \int_a^b w(x) f_2(x) f_j(x) dx + \cdots \\ + c_N \int_a^b w(x) f_N(x) f_j(x) dx = 0 \end{aligned}$$

Due to the orthogonality only one term remains [$w(x) \geq 0$]

$$c_j \int_a^b w(x) f_j^2(x) dx = c_j \lambda_j = 0$$

and therefore $c_j = 0$ for all j —the assumed relation does not exist.

The converse, that from a set of linearly independent functions we can construct an orthogonal set, can be shown by the Schmidt¹ process. Let the given set of linearly independent functions be $f_i(x)$. We compute

$$\int_a^b w(x) f_0^2(x) dx = \lambda_0 > 0 \quad w(x) \geq 0$$

¹ Eberhart Schmidt (1850–1890).

Then

$$g_0(x) = \frac{f_0(x)}{\sqrt{\lambda_0}}$$

defines the first orthonormal function $g_0(x)$. Using mathematical induction, we assume that we have constructed the first j orthonormal functions $g_i(x)$ ($i = 0, 1, \dots, j-1$). We set

$$F_j(x) = a_0 g_0 + a_1 g_1 + \dots + a_{j-1} g_{j-1} + f_j(x)$$

Now $F_j(x) \neq 0$, since the functions $f_i(x)$ are linearly independent and each $g_i(x)$ is a linear combination of the $f_k(x)$ for $k \leq i$. We need

$$\int_a^b w(x) F_j(x) g_i(x) dx = 0 \quad 0 \leq i \leq j-1$$

But this is, by the definition of $F_j(x)$,

$$\lambda_i a_i + \int_a^b w(x) g_i(x) f_j(x) dx = 0$$

which determines the a_i and hence $F_j(x)$. To normalize $F_j(x)$, we need to compute

$$\int_a^b w(x) F_j^2(x) dx = \lambda_j > 0 \quad w(x) \geq 0$$

and then set

$$g_j(x) = \frac{F_j(x)}{\sqrt{\lambda_j}}$$

Thus we have taken one more step in the induction.

If we have only a finite number of N samples x_m , then there are at most N linearly independent functions

$$f_j(x_m)$$

That there are N follows from the particular set

$$g_j(x_m) = \begin{cases} 0 & m \neq j \\ 1 & m = j \end{cases} \quad j = 1, \dots, N$$

since no set of these N functions $g_j(x_m)$ can be linearly dependent.

It is worth noting that in the construction process the set that we obtain is not unique unless we choose the particular $\sqrt{\lambda_i} > 0$ and also choose the set of linearly independent functions in a fixed order. Also, they obviously depend on the interval (a, b) and the weight function $w(x)$.

PROBLEMS

- 26.3.1 Given $P_0 = 1$, $P_1 = x$, and $w(x) = 1$, construct the orthogonal functions $P_2(x)$ and $P_3(x)$ over the set of points $-2, -1, 0, 1, 2$, where P_2 and P_3 are polynomials of degrees 2 and 3, respectively.
- 26.3.2 Given $1, \cos x, \cos^2 x$, and $w \equiv 1$, construct the orthogonal functions ($0 \leq x \leq \pi$).

26.4 LEAST-SQUARES FITS AND THE FOURIER COEFFICIENTS

In this section we show that least squares and orthogonality are closely related.

Theorem 26.4.1 The Fourier coefficients a_j give the best least-squares fit when a function $F(x)$ is expanded in terms of an orthonormal set of functions $g_j(x)$.

To prove this we want to minimize, for an arbitrary expansion,

$$\begin{aligned}
 m &= \int_a^b w(x) \left[F(x) - \sum_{j=0}^M c_j g_j(x) \right]^2 dx \\
 &= \int w(x) F^2(x) dx - 2 \sum_{j=0}^M c_j \int w(x) F(x) g_j(x) dx + \sum_{i=0}^M \sum_{j=0}^M c_j c_i \int w(x) g_i g_j dx \\
 &= \int w(x) F^2(x) dx - 2 \sum_{i=0}^M c_i a_i + \sum_{i=0}^M c_i^2 \\
 &= \int w(x) F^2(x) dx - \sum_{i=0}^M a_i^2 + \sum_{i=0}^M (a_i - c_i)^2
 \end{aligned}$$

But this is minimum only if each $c_i = a_i$ as required.

A remarkable and very useful property is that in the best least-squares fit with orthogonal functions each coefficient a_i is determined independently of all the others, and that if we decide to change the number of functions $g_i(x)$ we are using, then we need not redetermine any of the coefficients that we have already found. We therefore examine the converse problem.

If the coefficients c_i in a least-squares fit of a function $F(x)$ in terms of a set $\mu_i(x)$ are not to change as we change the number of $\mu_i(x)$ that we use, then the $\mu_i(x)$ must be orthogonal. Set

$$g(c_0, c_1, \dots, c_n) = \int_a^b w(x) \left[F(x) - \sum_{i=0}^M c_i \mu_i(x) \right]^2 dx$$

Since g is to be minimized,

$$\frac{\partial g}{\partial c_j} = 0 = -2 \int_a^b w(x) \left[F(x) - \sum_{i=0}^M c_i \mu_i \right] \mu_j(x) dx$$

or

$$\int w(x)F(x)\mu_j dx = \sum_{i=0}^M c_i \int_a^b w(x)\mu_i \mu_j dx$$

This must also be true for $M+1$ if the property is to be true for all M .

$$\int w(x)F(x)\mu_j(x) dx = \sum_{i=0}^{M+1} c_i \int_a^b w(x)\mu_i \mu_j dx \quad c_{M+1} \neq 0$$

Subtracting the previous equation, we get

$$c_{M+1} \int_a^b w(x)\mu_{M+1}\mu_j dx = 0$$

for any j ; that is, the μ_j are orthogonal to μ_{M+1} (and M was arbitrary).

Thus we see that orthogonal functions, the Fourier determination of the coefficients in an expansion, and the idea of least-squares approximation are all bound together.

26.5 BESSEL'S¹ INEQUALITY AND COMPLETENESS

The Fourier coefficients

$$a_j = \int_a^b w(x)F(x)g_j(x) dx$$

of an *orthonormal* set $g_j(x)$ satisfy the Bessel inequality

$$\int_a^b w(x)F^2(x) dx \geq \sum_{j=0}^M a_j^2 \quad w \geq 0$$

The proof is straightforward. We write, since the integrand is nonnegative,

$$\int_a^b w(x)[F(x) - \sum_{i=0}^M a_i g_i(x)]^2 dx \geq 0$$

and multiply out:

$$\begin{aligned} 0 \leq \int_a^b w(x)F^2(x) dx - 2 \int_a^b w(x)F(x) \sum_{i=0}^M a_i g_i(x) dx \\ + \int_a^b \sum_{i=0}^M \sum_{j=0}^M a_i a_j w(x) g_i g_j dx \end{aligned}$$

Using the definition of the a_i and the orthogonality of the g_i , we get

$$\int_a^b w(x)F^2(x) dx \geq 2 \sum_{i=0}^M a_i^2 - \sum_{i=0}^M a_i^2 = \sum_{i=0}^M a_i^2$$

for all M .

¹ Friedrich Wilhelm Bessel (1784–1846).

In the continuous case, if the equality holds for every continuous function $F(x)$ in the interval $[a, b]$ then the infinite set of functions is said to be "complete," and the equality

$$\int_a^b w(x) F^2(x) dx = \sum_{i=0}^{\infty} a_i^2$$

is called *Parseval's¹ equality*.

In the finite case we have, of course, no trouble in the process of interchanging limits to get Parseval's equality, and any set of N functions linearly independent over a set of N points is complete over the same set of points. Parseval's equality is an "ergodic theorem"; the sum of the squares of the function is the same as the weighted sum of the squares of the coefficients.

In Bessel's inequality the difference of the two terms is exactly the sum of the squares of the errors $\varepsilon(x)$ (residuals)

$$\sum_{i=1}^M w(x_i) \varepsilon_i^2 = \int_a^b w(x) F^2(x) dx - \sum_{j=1}^N a_j^2$$

In the discrete case the integrals become sums

$$\sum_{i=1}^M w(x_i) \varepsilon_i^2 = \sum_{i=1}^M w(x_i) F^2(x_i) - \sum_{j=1}^N a_j^2$$

PROBLEMS

26.5.1 If the functions are only orthogonal, then derive the corresponding formula for the sum of the squares of the ε_i .

26.6 ORTHOGONAL POLYNOMIALS

Orthogonal polynomials are a very important subclass of the class of orthogonal functions. They are defined to be such that the k th one ($k = 0, 1, 2, \dots$) is of degree exactly k , and they naturally have all the properties of orthogonal functions plus those peculiar to polynomials.

It is easy to show that the k th orthogonal polynomial $y_k(x)$ has exactly k real, distinct roots in the range of integration. Let us suppose that $r < k$. We form the product of the distinct, odd-multiplicity zeros

$$p(x) = (x - x_1)(x - x_2) \cdots (x - x_r) \quad r < k$$

¹ August von Parseval (1861–1942).

Then

$$\int_a^b w(x)p(x)y_k(x) dx = 0 \quad w(x) \geq 0$$

because $p(x)$ can be written as a sum of $y_0, y_1, \dots, y_r$, and $y_k(x)$ is orthogonal to all of them. But this is impossible since the integrand does not change sign in the interval. Hence there are k real, distinct roots in (a, b) .

The orthogonal polynomials $y_k(x)$ satisfy a three-term recurrence relation of the form

$$a_k y_{k+1}(x) + (b_k - x)y_k(x) + c_k y_{k-1}(x) = 0, \quad k \geq 1$$

To show this, we set

$$y_i(x) = \alpha_i x^i + \dots \quad \alpha_i > 0$$

(The choice of α_i as any particular positive number is a matter of convenience only.) Then

$$a_k y_{k+1} - x y_k$$

is a polynomial of degree k , provided that we choose $a_k = \alpha_k/\alpha_{k+1}$. Hence

$$a_k y_{k+1} - x y_k = \gamma_k y_k + \gamma_{k-1} y_{k-1} + \dots + \gamma_0 y_0$$

We multiply by $w(x)y_m(x)$ and integrate. We obtain

$$\int_a^b w(x)(a_k y_{k+1} - x y_k)y_m(x) dx = \gamma_m \lambda_m$$

For $m = 0, 1, \dots, k$, y_m is orthogonal to y_{k+1} ; and for $m = 0, 1, \dots, k-2$, $x y_m$ is a polynomial of degree less than k , and hence is orthogonal to y_k . Thus $\gamma_0 = \gamma_1 = \dots = \gamma_{k-2} = 0$. For $m = k-1$, the above equation becomes

$$\begin{aligned} - \int_a^b w y_k (x y_{k-1}) dx &= - \int_a^b w y_k \left(\frac{\alpha_{k-1}}{\alpha_k} y_k + c_{k-1} y_{k-1} + \dots \right) dx \\ &= - \frac{\alpha_{k-1}}{\alpha_k} \lambda_k = \gamma_{k-1} \lambda_{k-1} \end{aligned}$$

Using the three-term recurrence relation form, we see that

$$-c_k = \gamma_{k-1} = - \frac{\alpha_{k-1} \lambda_k}{\alpha_k \lambda_{k-1}} \neq 0$$

For $m = k$, the above equation becomes

$$\int_a^b w(a_k y_{k+1} - x y_k)y_k dx = - \int_a^b w x y_k^2 dx = \gamma_k \lambda_k = -b_k \lambda_k$$

Thus we can write

$$\frac{\alpha_k}{\alpha_{k+1}} y_{k+1} - x y_k = \gamma_k y_k + \gamma_{k-1} y_{k-1}$$

or

$$\frac{\alpha_k}{\alpha_{k+1}} y_{k+1} - (\gamma_k + x) y_k + \frac{\alpha_{k-1}}{\alpha_k} \frac{\lambda_k}{\lambda_{k-1}} y_{k-1} = 0$$

Several important results follow from this formula. We first show that the k zeros of $y_k(x)$ are separated by the $k - 1$ zeros of $y_{k-1}(x)$, provided that $w(x) \geq 0$. The proof is by induction. As a basis for the induction, we have $y_0(x) = \alpha_0 > 0$ and $y_1(x) = \alpha_1(x + \gamma_0)$. We know that $y_1(x)$ has a real zero in the range of integration, since

$$\int_a^b w(x) y_0(x) y_1(x) dx = 0 \quad \text{hence } a < -\gamma_0 < b$$

We now assume by the induction hypothesis that the zeros of $y_k(x)$ are interlaced by those of $y_{k-1}(x)$. At the zeros x_i of $y_k(x)$, the three-term relation becomes

$$\frac{\alpha_k}{\alpha_{k+1}} y_{k+1}(x_i) = - \frac{\alpha_{k-1}}{\lambda_k} \frac{\lambda_k}{\lambda_{k-1}} y_{k-1}(x_i)$$

At the top of the range ($x = b$) both $y_{k+1}(b)$ and $y_{k-1}(b)$ are of positive sign, since we chose the leading coefficients $\alpha_i > 0$ and since all the zeros of the functions lie inside the interval. At the largest zero of $y_k(x_i) = 0$, $y_{k-1}(x_i)$ is still positive; hence $y_{k+1}(x_i)$ is negative. At the next smaller zero, y_{k-1} has, owing to the induction hypothesis, changed sign; hence $y_{k+1}(x)$ is positive. And so it proceeds; as we pass from zero to zero of $y_k(x_i)$, $y_{k-1}(x_i)$ changes sign; hence $y_{k+1}(x_i)$ changes sign. Thus we have shown that the zeros of $y_k(x)$ interlace those of $y_{k+1}(x)$, except for the smallest zero of y_{k+1} . But since we have just shown that $y_{k+1}(x)$ has $k + 1$ real, distinct roots in (a, b) the last zero of $y_{k+1}(x)$ is smaller than all those of $y_k(x)$, and the induction proof is complete.

There is another important consequence of the three-term recurrence relations. Once we know the coefficients of the recurrence relation as functions of k and know $y_0(x)$ and $y_1(x)$, we can then compute the $y_k(x)$, one at a time, by this relation rather than by the cumbersome Schmidt process. See Sec. 27.2.

Orthogonal polynomials have another rather obvious property that we have already used, namely that an arbitrary polynomial $F_N(x)$ of degree N can be written as a sum of orthogonal polynomials $P_m(x)$ of degrees 0 through N . The

process for doing this is straightforward. We merely look at the coefficients of the highest powers both of the given polynomial $F_N(x)$ and of the corresponding orthogonal polynomial $P_N(x)$ of the same degree, and we form a suitable multiple $KP_N(x)$ of the orthogonal polynomial so that when this is subtracted from the given polynomial $F_N(x)$, the highest power cancels out. We then repeat this process on what is left, $F_{N-1}(x)$, which is a polynomial of lower degree. Enough steps get us to identically zero. Thus what we have subtracted is the orthogonal polynomial representation of the given polynomial. This is illustrated in the next section by using a particular set of orthogonal polynomials.

26.7 THE LEGENDRE¹ POLYNOMIALS

One of the most commonly occurring sets of orthogonal polynomials is the Legendre polynomials $P_n(x)$, which are the ultraspherical polynomials with $\alpha = 0$ (see Sec. 26.2). Thus we have

$$\int_{-1}^1 P_m(x)P_n(x) dx = \begin{cases} 0 & m \neq n \\ 2/(2n+1) & m = n \end{cases}$$

where we have not derived the normalizing factor $2/(2n+1)$. The three-term recurrence relation (Sec. 26.6) is

$$(n+1)P_{n+1}(x) - (2n+1)xP_n(x) + nP_{n-1}(x) = 0$$

This can be used to generate the Legendre polynomials (see Fig. 26.7.1), using

$$P_0 = 1$$

$$P_1 = x$$

$$P_2 = \frac{1}{2}(3x \cdot x - 1) = \frac{3x^2 - 1}{2}$$

$$P_3 = \frac{1}{3}\left(5x \frac{3x^2 - 1}{2} - 2x\right) = \frac{5x^3 - 3x}{2}$$

$$P_4 = \frac{1}{4}\left(7x \frac{5x^3 - 3x}{2} - 3 \frac{3x^2 - 1}{2}\right) = \frac{35x^4 - 30x^2 + 3}{8}$$

etc.

¹ Adrien-Marie Legendre (1752–1833).

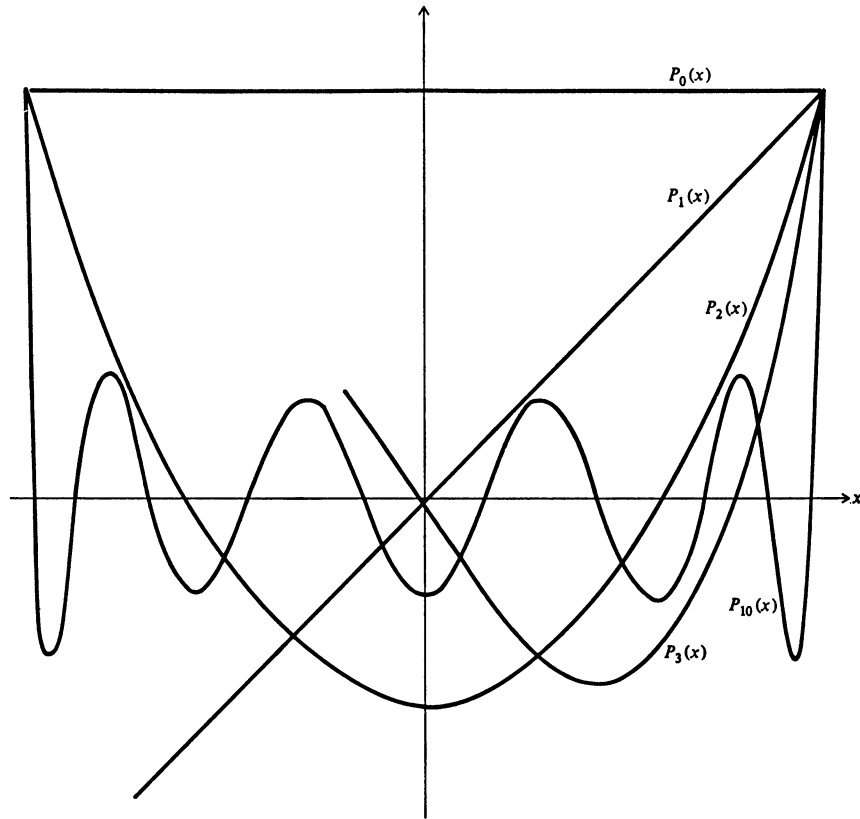


FIGURE 26.7.1

To convert a given polynomial, say

$$F(x) = 10x^3 + 3x^2 + 7x - 9$$

to a sum of Legendre polynomials, we subtract $4P_3$

$$10x^3 + 3x^2 + 7x - 9$$

$$4P_3 = 10x^3 \quad - 6x$$

$$\hline 3x^2 + 13x - 9$$

subtract

$$2P_2 = \quad 3x^2 \quad - 1$$

$$\hline 13x - 8$$

subtract

$$13P_1 = \quad -13x$$

and subtract

$$-8P_0 = \quad -8$$

Thus

$$F(x) = 10x^3 + 3x^2 + 7x - 9 \equiv 4P_3(x) + 2P_2(x) + 13P_1(x) - 8P_0$$

Other widely used sets of orthogonal polynomials are the Laguerre,¹ $L_n(x)$

$$\begin{aligned} \int_0^\infty e^{-x} L_m(x) L_n(x) dx &= \begin{cases} 0 & m \neq n \\ n! & m = n \end{cases} \\ (n+1)L_n(x) - (2n+1-x)L_n(x) + nL_{n-1}(x) &= 0 \quad L_0 = 1 \quad L_1 = 1-x \\ xL_n'' + (1-x)L_n' + nL_n &= 0 \end{aligned}$$

and Hermite² $H_n(x)$

$$\begin{aligned} \int_{-\infty}^\infty e^{-x^2} H_m(x) H_n(x) dx &= \begin{cases} 0 & m \neq n \\ 2^n n! \sqrt{\pi} & m = n \end{cases} \\ H_{n+1}(x) - 2xH_n(x) + 2nH_{n-1}(x) &= 0 \quad H_0 = 1 \quad H_1 = x \\ H_n'' - 2xH_n' + 2nH_n &= 0 \end{aligned}$$

Many other relations can be found among these functions in standard texts.

26.8 ORTHOGONAL POLYNOMIALS AND GAUSSIAN QUADRATURE

The orthogonal polynomials are connected with Gaussian quadrature (Chap. 19). To see this, consider again the problem of finding a formula of the form

$$\int_{-1}^1 f(x) dx = \sum_{i=1}^N w_i f(x_i)$$

where this is to be exact for all powers of x up to $2N-1$. Thus the formula should be true for an arbitrary polynomial $f_{2N-1}(x)$; that is,

$$\int_{-1}^1 f_{2N-1}(x) dx = \sum_{i=1}^N w_i f_{2N-1}(x_i)$$

Now consider dividing $f_{2N-1}(x)$ by the Legendre polynomial $P_N(x)$:

$$f_{2N-1}(x) = g(x)P_N(x) + r(x)$$

where the two functions $g(x)$ and $r(x)$ are polynomials of degree at most $N-1$. The integration formula becomes

$$\int_{-1}^1 g(x)P_N(x) dx + \int_{-1}^1 r(x) dx = \sum_{i=1}^N w_i g(x_i)P_N(x_i) + \sum_{i=1}^N w_i r(x_i)$$

¹ Edmund Nicolas Laguerre (1834–1886).

² Charles Hermite (1822–1901).

But since $P_N(x)$ is orthogonal to all lower Legendre polynomials, it is also orthogonal to *any* polynomial of degree less than N , and the first of the integrals vanishes.

On the other side of the equation we can get rid of the first summation if we pick the sample points as the zeros of the Legendre polynomial $P_N(x)$. We then have left the two terms

$$\int_{-1}^1 r(x) dx = \sum_{i=1}^N w_i r(x_i)$$

and we are reduced to determining the weights w_i in the standard case of known sample points x_i . From the uniqueness of the Gaussian quadrature the two results must be the same.

Notice how this parallels the earlier derivation which found the x_i so that the defining equations for $x^N, \dots, x^{2N-1}$ were satisfied, and then having found the x_i , we were reduced to finding the weights from the first N equations.

In an exactly similar way, the zeros of the Laguerre polynomials are the sampling points for the corresponding Gaussian quadrature formula

$$\int_0^\infty e^{-x} f(x) dx = \sum_{i=1}^N w_i f(x_i)$$

while the Hermite polynomials are related to the integration formula

$$\int_{-\infty}^\infty e^{-x^2} f(x) dx = \sum_{i=1}^N w_i f(x_i)$$

For this approach to Gaussian quadrature, see Mineur [41].

27.1 GENERAL REMARKS ON THE POLYNOMIAL SITUATION

The major use of the least-squares method is when the coefficients to be determined occur in a linear fashion, especially as coefficients of a polynomial. We shall therefore take up the polynomial situation first and then pass on to the other cases.

As was indicated in Sec. 25.6, the determinant of the normal equations tends to be very small; hence the solution for coefficients is likely to be uncertain. It is necessary to distinguish, however, between two things: the accuracy of the coefficients and the smallness of the sum of the squares of the errors. When the determinant is small, the coefficients may be poorly determined (that is, the curve is ill-defined by the data), but still the sum of the squares of the errors can be close to the minimum. Alternatively, it may be that the direct approach to finding the curve is at fault. Generally speaking, when there are up to five or six coefficients to be determined, the direct solution of the normal equations is usually satisfactory, but above this there is likely to be trouble.

The theory of Chap. 26 indicates that the expansion in orthogonal polynomials is a substitute for the direct solution. However, experience shows that if the orthogonalization is attempted by means of the Schmidt process, then the

same difficulty arises in a different disguise. In the Schmidt process, the construction of the m th polynomial is done by first subtracting all the components of the vector x^m , that is, $(x_1^m, x_2^m, \dots, x_N^m)$, which lie in the direction of the previously determined polynomials. The result is apt to be small when m is fairly large, and it is this remainder that is normalized as the last step in the Schmidt process. It is the cancellation that increases the errors, not the dividing factor which is usually much smaller than 1.

Loosely speaking, the trouble may be expressed by the remark that for large n and $x \geq 0$ the vector x^n points in about the same direction as does x^{n-1} , x^{n-2} , etc. Thus the equations leading to the Hilbert determinant are almost linearly dependent; hence the determinant is small and the accuracy low.

A novel and effective solution to this dilemma is offered in Sec. 27.3.

27.2 USE OF THE THREE-TERM RECURRENT RELATION

If we insist on using the orthogonal polynomials, then we can apparently avoid the troubles mentioned in the last section by using the three-term recurrence relation of Sec. 26.6 to generate the orthogonal polynomials.¹

We write the three-term recurrence relation in the form

$$\begin{aligned} p_0(x) &= 1 \\ p_1(x) &= xp_0(x) - \alpha_1 p_0(x) \\ p_{k+1}(x) &= xp_k(x) - \alpha_{k+1} p_k(x) - \beta_{k+1} p_{k-1}(x) \quad k \geq 1 \end{aligned}$$

where α_{k+1} and β_{k+1} are to be determined.

We first determine α_1 . We know that

$$\int_a^b w(x) p_0(x) p_1(x) dx = 0$$

Hence

$$\int_a^b w(x) x dx = \alpha_1 \int_a^b w(x) dx$$

We now suppose that we have $p_0(x)$, $p_1(x)$, $\dots$, $p_k(x)$ and that they are orthogonal to one another. We need to compute $p_{k+1}(x)$, the next polynomial of the set.

¹ G. E. Forsythe, Generation and Use of Orthogonal Polynomials for Data-fitting with a Digital Computer, *J. Soc. Ind. Appl. Math.*, vol. 5, no. 2, June, 1957; M. Ascher and G. E. Forsythe, SWAC Experiments on the Use of Orthogonal Polynomials for Data Fitting, *J. Assoc. Computing Machinery*, vol. 5, no. 1, January, 1958.

We require both

$$\int w(x)p_{k+1}(x)p_k(x) dx = 0$$

and

$$\int w(x)p_{k+1}(x)p_{k-1}(x) dx = 0$$

These suffice to determine α_{k+1} and β_{k+1} . Using the definition of $p_{k+1}(x)$, we get after some rearrangements,

$$\begin{aligned} \int w(x)xp_k^2(x) dx &= \alpha_{k+1} \int w(x)p_k^2(x) dx + \beta_{k+1} \int w(x)p_k(x)p_{k-1}(x) dx \\ \int w(x)xp_k(x)p_{k-1}(x) dx &= \alpha_{k+1} \int w(x)p_k(x)p_{k-1}(x) dx + \beta_{k+1} \int w(x)p_{k-1}^2(x) dx \end{aligned}$$

Since $p_k(x)$ and $p_{k-1}(x)$ are orthogonal, we have

$$\begin{aligned} \alpha_{k+1} &= \frac{\int w(x)xp_k^2(x) dx}{\int w(x)p_k^2(x) dx} \\ \beta_{k+1} &= \frac{\int w(x)xp_k(x)p_{k-1}(x) dx}{\int w(x)p_{k-1}^2(x) dx} \end{aligned}$$

The denominator of the β_{k+1} equation was computed during the previous step when $\alpha_k(x)$ was determined; thus we have three integrals to compute on each step.

That $p_{k+1}(x)$ is orthogonal to all $p_i(x)$ ($i < k-1$) follows from the equations

$$p_{k+1}(x) = xp_k(x) - \alpha_{k+1}p_k(x) - \beta_{k+1}p_{k-1}(x)$$

since, on multiplying by $w(x)p_i(x)$ and integrating, we have

$$\int w(x)p_k(x)xp_i(x) dx - \alpha_{k+1} \int w(x)p_k(x)p_i(x) dx - \beta_{k+1} \int w(x)p_{k-1}(x)p_i(x) dx = 0$$

The last two integrals are zero because of orthogonality, and in the first $xp_i(x)$ is a polynomial of degree less than k ; hence the integral is also zero.

When we try to construct polynomials over a discrete set of points x_j ($j = 1, \dots, N$), the integrals are replaced by sums. If we attempt to construct more than N polynomials [that is, above $p_{N-1}(x)$], then the equation determining α_N fails, as it should.

PROBLEMS

27.2.1 Orthogonalize $1, x, x^2, x^3$ over the interval $0 \leq x \leq 1$, using the three-term recurrence method.

27.3 THE CONSTRUCTION OF QUASI-ORTHOGONAL POLYNOMIALS

The direct approach to the least-squares fitting of polynomials leads to a system of normal equations which has large nonzero terms in places off the main diagonal and which is often hard to solve. The use of orthogonal polynomials leads to equations for which all the terms off the main diagonal are zero; hence the solution of the equations is trivial.

If we attempt to construct some orthogonal polynomials by the Schmidt process (or by the three-term recurrence relation) and do not determine them accurately, then the resulting equations to be solved have large terms on the main diagonal, while those off the diagonal are proportional to the deviation from orthogonality. If these are not large, then the system of equations is fairly easy to solve.

This observation suggests that we try our hand at constructing some *quasi-orthogonal polynomials* without doing much computation. Insofar as we make them orthogonal, the terms off the main diagonal of the system of normal equations will be zero, and insofar as we miss orthogonality, they will be that far away from zero.

Further meditating on this question suggests that the fundamental property we should try to keep is the interlacing zeros. Hence we choose $p_0(x) = 1$ as the first polynomial. The second we take as a straight line of convenient slope but with a zero near the median of the set of points. We next pick a quadratic whose zeros are on each side of the straight line. And so we proceed; at each step we choose the zeros of our next polynomial so that they are separated by the preceding polynomial and also leave room for the next polynomials. Unless we try to determine a polynomial whose degree is near that of the exact-matching polynomial through all the points (and this is rarely if ever the case) there is enough freedom to select the zeros at convenient places to make the polynomials have convenient coefficients and thus lighten the burden of subsequent calculation.

Experience with this method shows that it is very effective; usually the off-diagonal terms of the system of equations are fairly small, and the equations are easy to solve.

As a very simple example, where we have picked one unequally spaced point to avoid complete triviality (though quite commonly the data are equally spaced), consider data $x = -1, -1/2, 0, 1/2, 1, 2$.

We pick

$$P_0 = 1$$

$$P_1 = x$$

$$P_2 = (x + \frac{1}{2})(x - 1) = x^2 - \frac{1}{2}x - \frac{1}{2}$$

x	P_0	P_1	P_2
-1	1	-1	1
$-\frac{1}{2}$	1	$-\frac{1}{2}$	0
0	1	0	$\frac{1}{2}$
$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$
1	1	1	0
2	1	2	$\frac{1}{2}$

$$\begin{aligned}
\sum P_0^2(x_i) &= 6 \\
\sum P_0(x_i)P_1(x_i) &= -1 - \frac{1}{2} + 0 + \frac{1}{2} + 1 + 2 = 2 \\
\sum P_1^2(x_i) &= 1 + \frac{1}{4} + \frac{1}{4} + 1 + 4 = 6\frac{1}{2} = \frac{13}{2} \\
\sum P_0(x_i)P_2(x_i) &= 1 + 0 - \frac{1}{2} - \frac{1}{2} + 0 + \frac{5}{2} = \frac{9}{2} \\
\sum P_1(x_i)P_2(x_i) &= -1 + 0 + 0 - \frac{1}{4} + 0 + 5 = 3\frac{3}{4} = \frac{15}{4} \\
\sum P_2^2(x_i) &= 1 + 0 + \frac{1}{4} + \frac{1}{4} + \frac{25}{4} = \frac{31}{4}
\end{aligned}$$

The matrix of the unknowns (numbers from the quasi-orthogonal polynomials) is

$$\begin{pmatrix} 6 & 2 & \frac{9}{2} \\ 2 & \frac{13}{2} & \frac{15}{4} \\ \frac{9}{2} & \frac{15}{4} & \frac{31}{4} \end{pmatrix}$$

To judge the size the off-diagonal elements, we need to normalize the quasi-orthogonal polynomials. This amounts to multiplying the i th row and the i th column by $1/\sqrt{a_{ii}}$. What amounts to the same thing, we compare $(a_{ij})^2$ with $a_{ii}a_{jj}$:

$$\begin{aligned}
(a_{12})^2 &= 4 \leq 6 \cdot 13/2 = 39 \\
(a_{13})^2 &= 81/4 \leq 6 \cdot 31/4 = 186/4 \\
(a_{23})^2 &= 275/16 \leq (13/2)(31/4) = 806/16
\end{aligned}$$

While we did not discuss the iterative method of solving linear equations, it should be fairly clear that if the main-diagonal terms tend to dominate the off-diagonal terms, then the equations can be solved easily—if no other way than by solving the i th equation for x_i and iterating until convergence.

PROBLEMS

27.3.1 Examine the matrix of the quasi-orthogonal polynomials ($-1 \leq x \leq 1$) $w = 1$, $P_0 = 1$, $P_1 = x$, $P_2 = x^2 - 1/4$, $P_3 = x(x^2 - 9/16)$.

27.4 ON THE DEGREE OF THE POLYNOMIAL TO USE

A problem that arises regularly in least-squares fitting is what degree polynomial to use. The Bessel formula (see Fig. 27.4.1)

$$D^2(N) = \sum_{i=1}^M \varepsilon_i^2 = \sum_{i=1}^M F(x_i) - \sum_{j=1}^N \lambda_j a_j^2$$

gives a handy measure of the quality of the fit, and plotting this quantity as a function of the degree N of the polynomial being used gives a clue.

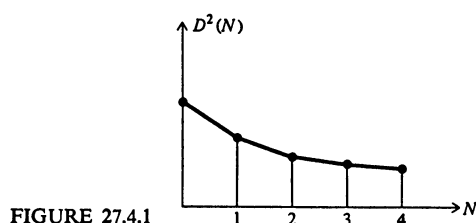


FIGURE 27.4.1

A better clue of when to stop in the process of fitting more and more terms in the least-squares approximations can be found from following the usual statistical practice of examining the residuals—in particular, examining the number of sign changes in the residuals.

Consider first the extreme condition of having only “noise” left in the residuals (see Fig. 27.4.2). Suppose that the residuals are random and independent of one another. By using simple probability we then expect that a given residual will be followed by one of the same sign half the time and one of the opposite sign half the time. Thus we expect to see about half the number of possible sign changes. If the number of sign changes is different from the number expected, either too low or too high by several times the square root of the number of possible changes, then we suspect that there was some kind of “signal” left and that our assumption of pure noise was wrong.

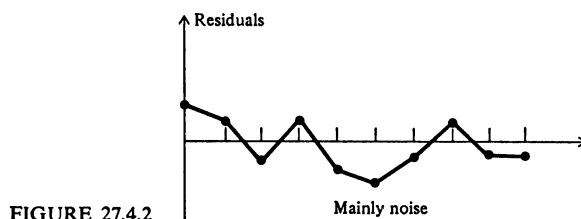


FIGURE 27.4.2

At the other extreme of no noise but all signal (see Fig. 27.4.3), then we expect (at least for the typical orthogonal system of functions) that the residuals will have about one more sign change than the last function we fitted, and a couple of extra sign changes above that will not worry us. The reason for this is that most orthogonal function systems have the property that the k th function has about k sign changes (this is exactly true of the important class of orthogonal polynomials; see Sec. 26.6).

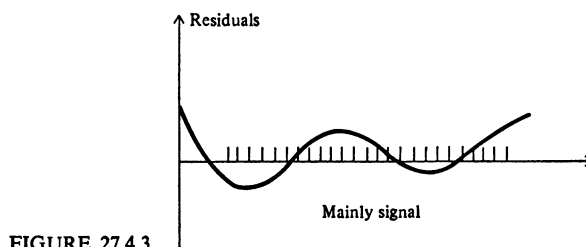


FIGURE 27.4.3

What we shall see in the residuals, of course, is probably a mixture of the two extremes, both some signal and some noise (see Fig. 27.4.4). The noise oscillations tend to increase the number of sign changes near the crossings due to the signal; and the more sign changes we see, the less we expect to reduce the sum of the squares if we fit a few more orthogonal functions.

Runs of constant sign often provide a more powerful test than the sign changes themselves—the runs reveal more structure. Furthermore, it is very easy on a computer to get the number of runs of each possible length by using an index register. If the index register is advanced one count each time the sign stays the same for the next point, and if when the sign changes the contents of the

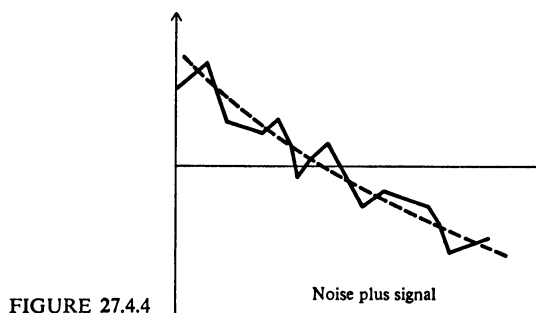


FIGURE 27.4.4

index register are used to determine the address whose contents are increased by 1, then the result is that the number of runs of length x are in location x (an initial count of 1 must be used for each run when the index register is reset after each use).

Thus we ask, What is the total length of the $k + 2$ longest runs? (We are assuming that we have fitted a polynomial of degree k and that the remainder will have $k + 1$ zeros.) If this total length is approximately the length of the entire interval, then we know that the crossings are confined to the neighborhoods of the zeros of the relatively noise-free remainder. If the intervals are broken up by noise, then the sum of the $k + 2$ longest intervals will not come near to the length of the interval.

This theory is a “down to earth” theory and is not given in the usual statistical courses. The test that is currently in statistical good graces is quite complex, has some dubious assumptions (all tests must make some assumptions), and probably is of less use to the practicing scientist (who is not highly trained in statistics and is therefore not in a position to appreciate what the orthodox test means). If you do understand advanced statistics, then by all means use it *if it seems appropriate*.

27.5 NONLINEAR PARAMETERS

It frequently happens that theory supplies the form that some data are to fit, and the data are to be used to determine the coefficients of the form, by the least-squares criterion. For example, suppose that the given form is

$$y(x) = a + be^{cx}$$

and we are to determine a , b , and c such that for the data (x_i, y_i) ($i = 1, \dots, N$)

$$\sum_{i=1}^N [y_i - (a + be^{cx_i})]^2 = \min = m \quad (27.5.1)$$

When we differentiate with respect to a , b , and c , respectively, and set the results equal to zero, the equations are difficult to solve.

Suppose that we plot the data and make some guess for the value $c = c_1$. We now determine the a and b in the usual way. We can then compute $m(c_1)$ from Eq. (27.5.1).

Next we try some other likely value of $c = c_2$ and again compute $m = m(c_2)$. An examination of these two values

c	$m(c)$
c_1	$m(c_1)$
c_2	$m(c_2)$

will suggest (by linear approximation) a new value of c , say c_3 .

In this way we can approach the minimum value of $m(c)$.¹ The details of the strategy for searching will be discussed in Chap. 43. It should be evident, however, that we shall use a great deal of machine computation as compared with the situation where *all* the parameters to be determined occur linearly. The more parameters that occur nonlinearly, the very much more the computation required to find the least-squares fit. Experience shows that when the number of nonlinear parameters reaches four or five, the process can be exceedingly painful and slow.

27.6 LEAST SQUARES WITH RESTRAINTS: CONTINUATION OF THE EXAMPLE IN SEC. 9.10

In the example in Sec. 9.10 we fitted an exact-matching polynomial to 11 data points $x = 0(0.1)1$, using Newton's interpolation formula. The success of the computation resulted in new laboratory equipment being built and much better and more extensive data being gathered.

A visual examination of the new data suggested about a sixth-order polynomial, and a plot of the logarithms of the values near $x = 1$ produced a slope near 6, thus tending to confirm the impression. With no further thought, a least-squares polynomial was computed, and the results were what one would expect—stupid, thoughtless computing produced foolish results! The polynomial began with a small positive value at $x = 0$, when the experiment clearly would give zero, and went slightly negative for a short distance, which was physically impossible.

The obvious remedy was to omit the constant term of the polynomial. Discussion with the physicist revealed that we could *not* expect the curve to be tangent to the x axis at $x = 0$, and so we could not leave out the x term. This more carefully considered computation produced an acceptable polynomial, and the rest of the computation was straightforward, as before.

Another obvious way of approaching the problem of fitting the data at $x = 0$ would have been to give the value there a large weight, say 1,000, where the other data points had weight 1.

Many variations occur in placing extra conditions on a least-squares fit. For example, suppose that we require the solution to take on two specified values, one at a and one at b , and are using orthogonal polynomials. We may use the classic method of Lagrange multipliers. If the values at a and b were zero, we

¹ We could try fitting a quadratic and using its minimum if we wished.

could use a weight function $(x-a)^2(x-b)^2$ and construct the corresponding orthogonal polynomials $P_n(x)$. Then the polynomials

$$Q_n(x) = (x-a)(b-x)P_n(x)$$

are orthogonal polynomials with weight factor $w(x) = 1$ passing through zero at $x = a$ and $x = b$.

PROBLEMS

- 27.6.1 Show that in the orthogonal-polynomial approach a straight line may be subtracted from the data first, to reduce two conditions $f(a) = A$ and $f(b) = B$ to $f(a) = f(b) = 0$.

27.7 SMOOTHING BY LEAST-SQUARES FITTING

Instead of using a single polynomial to fit all the data, it is often necessary to use a composite formula, as it were, and fit a least-squares polynomial *locally*. Typically an odd number of consecutive points is used to fit a low-order polynomial, and the value of this polynomial is taken to be the “smoothed value” at the middle point. For example, suppose we smooth by taking a least-squares fit on five equally spaced points, using a quadratic. We have, using the points $x = -2, -1, 0, 1, \text{ and } 2$,

$$f(a, b, c) = \sum_{i=-2}^2 (a + bx_i + cx_i^2 - y_i)^2$$

to minimize. The normal equations are

$$\begin{array}{l|l} 5a + 0b + 10c = \sum y_i & 17 \\ 0a + 10b + 0c = \sum x_i y_i & \\ 10a + 0b + 34c = \sum x_i^2 y_i & -5 \end{array}$$

Multiply the top equation by 17 and the bottom by -5 and add to find a , the value of the quadratic at the midpoint $x = 0$,

$$\begin{aligned} (85 - 50)a &= 17 \sum y_i - 5 \sum x_i^2 y_i \\ a &= \frac{1}{35} \sum_{i=-2}^2 (17 - 5x_i^2)y_i \\ &= \frac{-3y_{-2} + 12y_{-1} + 17y_0 + 12y_1 - 3y_2}{35} \end{aligned}$$

Thus we have a formula for smoothing the data (except for the four end values). To get formulas for the end values, we have to find both b and c and evaluate the quadratic at $x = \pm 1$ and ± 2 .

PROBLEMS

27.7.1 Find the formulas for the four end values.

27.7.2 Carry out the smoothing for a straight line $y = a + bx$.

Ans. $\frac{1}{4}(y_{-2} + y_{-1} + y_0 + y_1 + y_2)$

27.7.3 Smooth, using seven equally spaced points and fitting a straight line, a quadratic and a quartic.

27.7.4 Discuss the fitting of an even-degree polynomial for the above method of smoothing.

27.8 ANOTHER FAULT OF LEAST-SQUARES FITTING

We have already discussed (Sec. 25.2) the fault that one or more extreme values, quite likely gross errors, play a dominant role in shaping the least-squares function that is selected from the family of functions.

Another fault with polynomial least squares is that the error curve tends to have very large errors at the ends of the interval. This comes about because it is in the nature of polynomials to go to infinity for large values of the argument, and therefore the error is likely to be large at the ends. Indeed, supposing for the moment that the expansion in Legendre polynomials is rapidly converging, then the error for some finite degree is approximately the first term neglected and will therefore look like the $(N + 1)$ st Legendre polynomial (even if the Legendre polynomials are not used in the fitting process). A study of the plot of the first few polynomials given in Sec. 26.7 shows this end effect clearly. The remedy for this effect is to weight the end values relatively heavily; indeed, the correct weighting in a certain sense is given by the Chebyshev polynomials of the next two chapters.